

Taxes Depress Corporate Borrowing: Evidence from Private Firms

Ivan T. Ivanov

Federal Reserve Bank of Chicago, USA

Luke Pettit

George Mason University, USA

and

Toni M. Whited

University of Michigan and NBER, USA

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We use variation in state corporate income tax rates to re-examine the relation between taxes and corporate leverage. Contrary to prior research, corporate leverage rises after tax cuts for small private firms. An estimated dynamic equilibrium model shows that tax cuts make capital more productive and spur the use of leverage. Tax cuts also produce more distant default thresholds and lower credit spreads. These effects outweigh the lower interest tax deduction and lead to higher optimal leverage choices, especially for firms with flexible investment policies. The presence of the interest tax deduction raises consumer welfare in equilibrium.

Key words: Corporate leverage, Taxes

JEL codes: G31, G32

1. INTRODUCTION

Do taxes boost or depress corporate leverage? This old question has been at the center of corporate finance at least since (Modigliani and Miller, 1963), which emphasizes that the benefit from the interest tax deduction is valuable and likely larger than any countervailing costs of debt related to financial distress. This trade-off has received much attention in part because it informs corporate managers' decisions on financing capital and labour. More broadly, understanding the interaction between taxes and debt is of fundamental interest, as corporate debt can affect the pass-through of monetary policy to the real economy (Ottonello and Winberry, 2020), stock market risk (Gomes and Schmid, 2010), and macroeconomic stability

(Brunnermeier and Sannikov, 2014). As surveyed by Fleckenstein *et al.* (2019), most prior empirical work finds a positive relation between leverage and taxes, so the question appears to be settled and in line with the idea that firms value the interest tax deduction.

We tackle this old question in two new ways and find new empirical and quantitative answers. First, using data on private U.S. firms, we study changes in state corporate income taxes since the 1990s. We often find a negative relation between taxes and leverage, especially for small private firms, despite the tax deductibility of interest during our sample period. We also find anticipation effects, as firms adjust capital structure in response to the enactment of tax laws, which are often effective 1–2 years later.

While at first surprising, these results are more than simply contradictory, given that most prior research uses samples of large public companies (Titman and Wessels, 1988; Graham, 1996; Faccio and Xu, 2015; Heider and Ljungqvist, 2015). Instead, we focus on private firms, which face different trade-offs than their public counterparts, as they are less well-capitalized and face more financial frictions (Brown *et al.*, 2021). Moreover, examining private firms expands the scope of this earlier research in an interesting direction. For example, over half of new job creation in the U.S. occurs in privately held companies (Smith, 2007), and most world economic activity occurs in smaller, bank-dependent, privately held companies.

Second, to rationalize our results, we estimate a dynamic model of firm leverage. The intuition from the model is based on the simple idea that the tax benefit of debt is offset by two tax-sensitive costs that are not necessarily small. First, tax cuts raise firm profits and thus lower the likelihood of default. Credit spreads fall, so firms have an incentive to raise leverage. Second, a tax cut boosts the marginal product of capital. Because external equity finance is either costly or unavailable, if a firm lowers leverage in response to the lower tax benefit, it incurs an opportunity cost in terms of the lost marginal product, so the firm optimally substitutes into capital by using more leverage. Interestingly, Hundsdoerfer (2023) finds results consistent with these mechanisms. He studies a change in a VAT tax that leaves the interest tax deduction intact, finding that an increase in the VAT tax reduces leverage.

Turning to the details of our analysis, our primary data come from the Federal Reserve's Y-14 Collection, which is a supervisory data set that covers privately held, bank-dependent firms in the United States since 2011. The data set contains not only loan details but also firm-level accounting information. Using the Y-14 data to conduct simple event studies, we investigate how state corporate income tax cuts affect leverage. We focus on tax cuts because tax hikes seldom occur during our sample period. We use both the enactment and effective dates for the tax changes, as they can be signed into law 1–2 years before becoming effective. As such, an ancillary contribution of our work is the collection of tax enactment dates.

We find that small firms' leverage first responds to the tax cuts in the enactment year, with no significant reaction before enactment. The event study coefficients on or after the enactment dates are precisely estimated, ranging from approximately 4% in the enactment year to about 2% 4 years later. This result indicates the presence of anticipation effects, and the persistence of the results also points to a permanent increase in leverage in response to tax cuts. We also show that changes in long-term debt primarily drive this rise in leverage. Moreover, we find that bank assessments of loan default rates fall by 20–30 basis points after tax cuts. This last result is important because it supports the general notion from our model that default risk and the cost of debt are also tax-sensitive.

While small firms react to tax cuts, large firms do not, as we find no significant leverage changes for large firms after corporate income tax cuts. For our sample of public firms, we also find that leverage rises slightly in response to tax cuts, although effects do not persist.

We extensively examine the robustness of the event study results. We confront the biases that can arise in settings with heterogeneous treatment effects

(de Chaisemartin and D'Haultfoeuille, 2020). As in Romer and Romer (2010) or Giroud and Rauh (2019), we use narrative techniques to ensure tax cuts are not endogenously related to local economic activity. We also examine alternative thresholds to define firm size or corporate taxation status. In all cases, our results remain similar to those from our main specification. Our results are also similar when we exclude firms with access to private equity markets or financially distressed firms, when we expand or contract the list of control variables, or when we account for the timing gap between effective and enactment dates.

To provide intuition for these somewhat unusual results, we use our data to estimate a dynamic equilibrium model of leverage and investment in the spirit of Hennessy and Whited (2005, 2007). In this class of infinite-horizon models, firms choose investment and debt to maximize the expected discounted value of distributions to shareholders. They also receive an exogenous interest tax deduction. In our model, firms face an exogenous stochastic tax rate, so they optimize their policies considering the possible future evolution of corporate taxes, as in Hennessy and Strebulaev (2020). This tax benefit is offset by two endogenous tax-sensitive costs. First, because debt need not be fully collateralized, firms can default, which occurs when they have insufficient internal resources to repay their debt. A higher tax rate exerts downward pressure on optimal debt because credit spreads rise, as both default thresholds and lender recovery fall. Second, firms optimally substitute between debt and capital as a means of transferring resources through time. Because debt can be thought of as a (negative) linear storage technology, when capital becomes more productive, firms want to use capital relative to this less productive technology, so capital rises and negative debt falls; that is, debt rises.

Ex ante, it is not clear which of these effects is quantitatively more important. To understand this issue, we estimate the model parameters using our samples of large and small private firms. Our intent in this sample split is roughly to separate firms that face more or fewer financial frictions (Hadlock and Pierce, 2010). Among other moments, we target the sensitivity of leverage to a dummy for a tax cut. We find that we can match the positive sign in our sample of small firms and the marginally negative sign in our sample of large firms. One difference between these two estimations is the size of the capital adjustment costs parameter, which is several times larger for the sample of large firms than for the sample of small firms. High adjustment costs weaken the substitution between debt and capital, so the effect of the tax benefit is relatively more important. The other difference is the magnitude of default costs, with small firms having higher costs. In this case, productive firms stay closer to the default threshold, where the tax sensitivity of the cost of debt matters.

We use our model to quantify the welfare and misallocation effects of the tax benefit of debt. We compare two economies, both with corporate taxation, but one with an interest tax deduction and one without. We find a result with a second-best flavour: welfare is higher in the economy with a tax deduction. Without the deduction, firms find debt less attractive and hoard cash to offset the lack of external equity finance. This siphoning of resources into cash results in lower output, which, in equilibrium, results in a lower equilibrium real wage, more hours worked, and lower consumption and welfare. Misallocation also rises when we remove the tax benefit because the tax benefit boosts the use of debt to finance capital and offsets the adverse welfare effects of other frictions that shape corporate financing.

Finally, we examine alternative frictions that affect firms' leverage decisions. First, in models with collateral constraints, the cost of debt is lost financial flexibility, which is tax-sensitive. Second, in our baseline model, firms cannot issue external equity. We relax this assumption by allowing the existence of costly equity. Modest equity issuance costs imply that equity issuance is rare, so the tax-sensitive cost of debt continues to matter. Third, in our model, firms default when they have insufficient liquidity to repay debt. If we alter the model to allow firms to roll over debt unless they become insolvent, they are naturally less likely to default, but the

tax-sensitive cost of debt is still present. Fourth, in our data, taxes often become effective after they are enacted. To embed this feature of tax legislation in our model, we add a news shock (Jaimovich and Rebelo, 2009) to the model, finding a similar effect of tax news on leverage.

Our model builds upon a large literature in corporate finance that uses dynamic models of finance and investment to understand corporate leverage policy (e.g. Hennessy and Whited, 2005; Wu, 2018; Gao *et al.*, 2021). Our model is intentionally standard, containing no new ingredients designed specifically to generate the seemingly counterintuitive result of a negative sensitivity of leverage to corporate taxes. In other words, this result was already hiding in models of this class. Interestingly, our result has a related precedent in Hines and Park (2019). In this model, distortionary subsidies that favour one type of capital over another reduce firm productivity. Lenders then raise borrowing costs in response, and this effect can lower overall investment, even though spending on tax-favoured capital goods might rise.

We organize the paper as follows. Section 2 describes our data, Section 3 describes our empirical methodology, and Section 4 presents our results. In Section 5, we outline our model and present the estimation results and counterfactuals. Section 6 concludes.

2. DATA

2.1. *Data on corporate borrowing*

Our firm-level income statement and balance sheet data come from Schedule H.1 of the Federal Reserve's Y-14Q data collection, which started in June 2012 to support the Federal Reserve's bank stress tests (Board of Governors of the Federal Reserve System, 2022). These data are quarterly, and the sample period runs from the third quarter of 2011 to the present. The dataset contains information on the loan portfolios of the 33 largest banks in the U.S., which originate approximately three-quarters of all U.S. commercial and industrial lending. The Y-14Q data contain borrower financial statements, which borrowers provide to satisfy loan collateral and covenant requirements. Banks provide loan-level data on all commercial and industrial loans in their portfolio whenever a loan exceeds \$1 million in the commitment amount, together with the most recent financial statement information of the associated borrower, if available. The smallest companies in the Y-14 collection do not have financial statement data, likely because it is too costly, so they substitute tax returns, which we do not observe. Given our focus on state corporate tax changes in the U.S., we restrict the sample to all loans to U.S. addresses. While these data are quarterly, most firms only report financials every four to six quarters. So, we keep the financial statement information with a reporting date closest to the end of each calendar year, typically the information from the fourth quarter for the previous twelve months. Because some of the entities responsible for debt repayment can be subsidiaries, we merge the entities in the Y-14 Collection into parent firms using data from the National Establishment Time-Series (NETS) Database (Walls & Associates, 2020) and the Business Entity and Company Family Tree of Standard & Poor's (S&P Global, 2024). These two data sets span the period from 1990 to 2017.

We restrict the sample to domestic private borrowers, excluding foreign and U.S. state and local government entities, tribal governments, special purpose vehicles, individuals and private households, and other nonprofit organizations. We exclude public firms, as we study those separately using Compustat data (S&P Global, 2020). We require each borrower to report a zip code, which we use to identify the borrower's location, using the zip-county crosswalk distributed by the U.S. Department of Housing and Urban Development (United States Department of Housing and Urban Development, 2021). We also exclude firms that have loans guaranteed by U.S. government agencies, that are in default (non-accrual status), or that borrow through establishments or subsidiaries located in multiple states in the same quarter. This last restriction mitigates

the issue that firms are taxed based on their nexus in individual states, so they might not be particularly sensitive to a change in the tax rate in their home state if their revenues are dispersed across states. We also omit financial, regulated, and government entities, as outlined in [Online Appendix A](#). See [Brown *et al.* \(2021\)](#) for a more detailed description of the data set.

Although we do not observe firm tax-filing status, the small size of the companies in the Y-14 data suggests that a significant portion of firms are subject to individual taxation of pass-through income. For example, half of the firm years with financial statements in our data have less than \$19 million in total assets, and three-quarters have less than \$68 million in total assets. Because we focus on corporate income taxes, we screen for pass-through entities such as sole proprietorships, partnerships, and S corporations using the following restrictions. First, we exclude borrowing entities if the obligated entity or the guarantor of the debt is classified as “Individual” or “Sole Proprietor.” Second, we exclude firm-year observations in which book assets fall below \$100 million in the previous year. We impose this restriction because larger companies are much more likely to benefit from choosing corporate taxation and organizing as C corporations. While pass-through entities have a lower tax burden relative to C corporations, [Smith *et al.* \(2022\)](#) show that other aspects of organizing as a C-corporation are more important for large firms with more than \$100 million in sales; for example, no restriction on the number of shareholders or types of ownership, a lower tax burden on retained earnings, and a wider range of deductions. Imposing this restriction is likely conservative, as the Joint Committee on Taxation reports that virtually all assets of C corporations belong to those companies whose total assets exceed \$100 million ([Joint Committee on Taxation, 2024](#)).

For an observation to remain in the sample, we require the availability of the book value of total assets. We drop observations where total assets or net sales are equal to zero or negative, as well as observations with negative cash balances, current assets, current liabilities, depreciation, interest expense, tangible assets, or fixed assets. We also require the beginning-of-period book value of total assets to be available, as we use this variable to scale all financial variables. The resulting sample has 39,363 non-singleton firm-year observations from 2011 to 2017, where we truncate the end of the sample because we examine the dynamic effects of tax changes, as explained in Section 3, below. We winsorize all financial statement variables at the 1st and 99th percentiles to mitigate the effect of extreme observations.

While the Y-14 data provide some detail on firm-level accounting variables, its time period is limited. Therefore, we also use the Shared National Credit (SNC) data, which spans 1992 through the present ([Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency and Federal Deposit Insurance Corporation, 2020](#)). While the SNC dataset has a longer time series, with detailed data on loan commitment amounts, maturity, and credit utilization, it has the disadvantage that it does not contain firm-level balance sheet data.

The SNC program covers all syndicated deals that meet two criteria: they exceed \$20 million (\$100 million since 2018), and they are held by three or more unaffiliated institutions supervised by the Federal Reserve System, Federal Deposit Insurance Corporation, or the Office of the Comptroller of the Currency (OCC). Before 1999, this second inclusion criterion required deals to have at least two supervised unaffiliated lenders. The loan deals in this data collection comprise nearly the entire syndicated loan market in the U.S. in terms of loan amounts. We aggregate loan commitment amounts to the borrower level, and we include both the amounts of credit line sizes and term loan commitments. We restrict this sample to domestic firms and exclude government entities, utilities, financials, public administration entities, firms that have defaulted on their debt, firms that are either in non-accrual status or have “troubled-debt” restructurings, and religious institutions (in terms of either 2-digit NAICS or the SNC industry definitions). The resulting sample contains 50,203 non-singleton firm-year observations from 1993 to 2018 that

have at least 2 consecutive years of data and over \$100 million in loan commitments as of the prior year.

Our sample of public firms is from the CRSP-Compustat Fundamentals Annual database ([Center for Research in Security Prices, 2020](#)). Our sample period starts in 1989 to conform to the sample in [Heider and Ljungqvist \(2015\)](#) and ends in 2017 to be comparable with the sample of firms from the Y-14 data. Because a firm's headquarters location in Compustat is backfilled to the most recent location, we use a file provided by Standard & Poor's that contains the correct historical headquarters locations since 1994 ("CST_HIST," [S&P Global, 2020](#)). We exclude utilities, financials, and public administration entities (2-digit SIC code 49 and 1-digit SIC codes 6 and 9). We also exclude foreign firms (identified as having historical headquarters or incorporation locations outside of the U.S.), firms with missing headquarters locations, firms with negative or missing total assets, and firms with missing pre-tax earnings. The resulting sample has 91,314 non-singleton firm-year observations. The variable definitions for the CRSP-Compustat, SNC, and Y-14 samples are in [Online Appendix A](#).

2.2. State taxation and economic data

To identify the effective dates of state corporate taxes changes, we use the dataset on top statutory state corporate income tax rates since 1975 and other tax base characteristics assembled by [Serrato and Zidar \(2018a\)](#).¹ Although our data start in 1989, we use earlier tax changes to help calibrate the tax process for our model. For each tax change, we collect the corresponding enactment date, that is, the date the tax change became law. These two dates need not be the same. As an extreme example, Indiana approved a tax package in 2011 that lowered corporate income taxes from 8.5% to 6.25% between 2013 and 2017 (a 0.5% reduction in 2013, 2014, 2015, and 2016 and a 0.25% reduction in 2017). In this example, the Indiana tax package enters our sample only once, with 2013 as the effective year and 2011 as the enactment year. Out of the 125 tax legislation packages since 1975, 79 became effective in the same year or retroactively, 33 became effective in the following year, and 13 became effective in 2 or more years. For several of the packages that are effective immediately, their rates gradually increase or decrease for up to 5 years.

We identify all corporate income tax changes since 2012 from [The Tax Foundation \(2020\)](#) and gather the corresponding enactment dates from the statutory compilation of all states ([LexisNexis, 2020](#)). We collect data on all enactment dates prior to 2012 from amendments to the states' tax statutes. Specifically, we obtain electronic copies/scans of the tax statutes from each state's legislature/legal library. We then read through the statutes to identify the relevant corporate income tax rate changes and record the respective enactment dates, typically the date the state's governor signs the legislation.

Finally, we gather data on the effective dates of state sales tax rates and individual income tax rates between 2000 and 2020 from the [Tax Policy Center \(2023\)](#) and [The Federation of Tax Administrators \(2020\)](#), respectively, as well as information on Census Bureau Regions and Divisions from [U.S. Census Bureau, Population Division \(2017\)](#). We obtain annual data on state and county GDP as well as county population, employment, and per-capita income from the U.S. Bureau of Economic Analysis ([United States Bureau of Economic Analysis, 2019a, 2019b, 2020](#)). We also obtain monthly data on state unemployment rates from the U.S. Bureau of Labor

1. [Serrato and Zidar \(2018b\)](#) collect these data from a variety of sources including [Council of State Governments \(2011\)](#), [CCH Incorporated \(2017\)](#), [Bernthal et al. \(2012\)](#), [Chirinko and Wilson \(2008\)](#), and [Wilson \(2009\)](#). For more details on the construction of this dataset, please refer to [Serrato and Zidar \(2018b\)](#).

TABLE 1
Summary statistics

	Mean	St. Dev.	p10	p25	p50	p75	p90
Panel A: Y-14 Data							
Book Assets, \$m	2,151	37,758	124	166	288	681	1,934
Net Income	0.07	0.13	−0.03	0.01	0.05	0.09	0.16
Net Sales	1.82	1.47	0.39	0.78	1.46	2.43	3.64
EBITDA	0.14	0.15	0.03	0.07	0.11	0.18	0.26
Long Term Debt	0.31	0.28	0.00	0.07	0.26	0.49	0.70
Total Debt	0.38	0.28	0.01	0.15	0.35	0.57	0.76
Fixed Assets	0.32	0.28	0.03	0.09	0.25	0.49	0.76
Panel B: Compustat Data							
Book Assets, \$m	3514	7722	35	135	633	2615	8991
Net Income	−0.06	0.30	−0.37	−0.06	0.03	0.07	0.12
Sales	1.01	0.77	0.21	0.48	0.83	1.36	1.99
EBITDA	0.03	0.27	−0.28	0.03	0.10	0.16	0.22
Long Term Debt	0.20	0.21	0.00	0.00	0.15	0.32	0.49
Total Debt	0.23	0.23	0.00	0.01	0.19	0.36	0.54
Fixed Assets	0.24	0.24	0.02	0.06	0.15	0.34	0.63
Panel C: SNC Data							
Commitment Amount, \$m	689	1,136	125	180	325	725	1516
Utilized Amount, \$m	283	588	0	28	122	290	662
Utilization Ratio	0.48	0.35	0.00	0.09	0.51	0.80	0.95
Contract Maturity, months	68	26	37	56	61	79	97
Panel D: Characteristics of Tax Changes							
Tax Cuts (N = 50)							
Rate Change, %	−1.22	1.43	−2.25	−1.5	−0.72	−0.4	−0.22
Initial Rate, %	6.89	2.18	4.44	5.81	7.05	8.25	8.91
Resulting Rate, %	5.67	2.81	3.3	4.5	6.12	7.36	8.19
Tax Hikes (N = 24)							
Rate Change, %	1.39	1.14	0.5	0.71	1.02	1.5	2.8
Initial Rate, %	7.93	1.85	6.0	6.5	7.95	9.0	9.78
Resulting Rate, %	9.31	2.41	7.0	7.99	8.7	10.2	12.61

Notes: This table presents summary statistics for our sample of private firms from the Y-14 data (Panel A), our sample of public firms from Compustat (Panel B), firm-level loan characteristics from the SNC data (Panel C), and the tax changes used in our analyses (Panel D). The commitment amount refers to the sum of credit line and term loan commitments, while the utilized amount is the sum of term loans and the outstanding drawn amounts under credit lines. All firm financial variables in Panels A and B, with the exception of Book Assets, are scaled by lagged firm total assets.

Statistics ([United States Bureau of Labor Statistics, 2024](#)), daily data on Aaa and Baa corporate bond yields from Moody's ([Moody's, 2024a, 2024b](#)).

2.3. Descriptive statistics

We describe our sample of firms from the Y-14 data in Panel A of Table 1. The median company has \$288 million in book assets, and the interquartile range is from \$166 to \$681 million. In addition, sample firms have total debt-to-assets of approximately 38% and long-term debt-to-assets of 31%. Both measures of leverage exhibit substantial variability. In comparison, as seen in Panel B, our sample of public firms from Compustat contains firms that are, on average, much larger, less leveraged, and less profitable. In addition, public firms have fewer fixed assets relative to their size, on average.

[Online Appendix Figure B1](#) presents the distributions of size and leverage of the Y-14 and Compustat firms. While both samples contain a large fraction of firms with assets less than

\$500 million, there are many more very large firms in Compustat. Next, while both samples contain substantial fractions of firms with zero leverage, the Y-14 sample contains a much larger fraction with levels of elevated leverage.

Panel C of Table 1 describes the sample of firms from the SNC. Most of the firms in this sample are private and obtain most of their external financing through bank borrowing. We observe that these firms have a significant amount of slack in their credit lines. The median amount of loan commitments is \$325 million, but for the median firm-year, we see only about 51% of these commitments used.

Panel D summarizes the state corporate income tax hikes and tax cuts that became effective from 1987 to 2017. The 74 tax legislations constitute 24 tax hikes and 50 tax cuts. Tax hikes increase corporate income taxes by an average of 1.39%, while tax cuts reduce taxes by about 1.22%, where these figures include the cumulative effect of staggered tax legislation. Prior to the effective date of tax changes, state corporate income tax rates are higher in states with subsequent tax hikes than in states with tax cuts. For example, initial corporate income tax rates average 7.93% in tax-hike states and 6.89% in tax-cut states.

3. EMPIRICAL APPROACH

We estimate the effects of tax changes on firm leverage with difference-in-differences event study regressions of the form:

$$y_{it} = a_i + b_{mt} + \sum_{k=-2}^{k \geq +4} \bar{c}_k \mathbf{1}\{K_{it} = k\} + \gamma X + \epsilon_{it}, \quad (1)$$

where i , m , t , and k denote firms, industries, years, and years relative to the event of interest, respectively. Dates preceding the event correspond to $k < 0$, and $k \geq 0$ corresponds to dates after the event, with $k \geq +4$ representing 4 or more years after the event. The dummy variables $\mathbf{1}\{K_{it} = k\}$ are leads and lags of the event indicator, so the corresponding dynamic treatment effects are given by $\bar{c}_k$. The outcome of interest is given by y_{it} , X is a vector of control variables described below, and $\bar{a}_i$ and $\bar{\beta}_{mt}$ are firm and industry-year fixed effects. Given the inclusion of firm fixed effects, $\bar{a}_i$, the event study estimates represent deviations from the average level of the outcome of interest for a given firm. As noted in [Borusyak et al. \(2024\)](#), this specification is valid under the assumption that the treatment effects are homogeneous across units and calendar time, depending only on k .

The model in equation (1) is identified up to a linear trend ([Borusyak et al., 2024](#)). One identifying assumption is the specification of a base year before which no pre-trends are present. We define this omitted category as years $k \leq -3$ relative to the event. We choose this lag length because tax legislation can be enacted 1 or 2 years before becoming effective. If firms react to enactments, then in our analysis of tax-effective dates as events, the lagged treatment effects might enter (1) with significant coefficients, so an omitted category greater than -3 would result in omitted variable bias.

Equation (1) is preferable to a standard difference-in-differences specification, such as:

$$y_{it} = \bar{a}_i + \bar{\beta}_t + cD_{it} + \gamma X + \epsilon_{it}, \quad (2)$$

where D_{it} is the interaction term dummy variable. Although the estimator in equation (1) is less efficient than that in (2) ([Borusyak et al., 2024](#)), equation (2) is only valid under the restrictive assumption that the $\bar{c}_k$'s in equation (1) are all equal for $k > 0$. This requirement means that the treatment leads to an immediate and permanent jump in the outcome variable and no further

effects. If this assumption is violated, the estimate of c in equation (2) is biased and difficult to interpret because c is the weighted average of $\tilde{c}_k$'s in equation (1), and not all the weights need to be positive (de Chaisemartin and D'Haultfoeuille, 2020; Goodman-Bacon, 2021; Borusyak *et al.*, 2024). This work explains that bias in c stems from using post-treatment periods to provide counterfactuals for earlier periods.

The vector of controls, X , differs for each of our three datasets because of data limitations. For the regressions using the Y-14 data, we include firm internal credit rating fixed effects, which range from AAA to D, and the log of lagged firm sales. We include firm and 4-digit NAICS industry-by-year fixed effects. For our regressions using Compustat data, we include firm and 4-digit SIC industry-by-year fixed effects, as well as lagged levels of log total assets, the ratio of net income to assets, the ratio of property, plant, and equipment to total assets, and the market-to-book ratio. For the regressions using either Compustat or the SNC data, we include lagged levels and changes in the state unemployment rate and log GDP. For the Y-14 regressions, we also include county-level lagged levels and changes in log employment, per-capita income, log population, and log GDP.

Several controls are common across all three data sets. Following Serrato and Zidar (2018b), we control for the structure of the corporate tax base using the fourteen following measures: an indicator for throwback rules, an indicator for combined reporting rules, an indicator for whether the R&D tax credit applies to an incremental base that is a moving average of past expenditures, an indicator for whether the R&D tax credit applies to an incremental base that is fixed on a level of past expenditures, the number of years for loss carrybacks, number of years for loss carryforwards, an indicator for franchise taxes, an indicator for federal income tax deductibility, an indicator for the federal income tax base as the state tax base, an indicator for federal accelerated depreciation, an indicator for accelerated cost recovery system (MACRS) depreciation, an indicator for federal bonus depreciation, and sales and payroll apportionment weights. We extend these tax base measures through 2017 with information from CCH Incorporated (2017) and LexisNexis (2020). Our final control is the top statutory state individual income tax rates, which we obtain from The Federation of Tax Administrators (2020).

One potential concern with our tests is that contemporaneous changes in state individual income taxes might drive corporate leverage. For example, our results could be attributed spuriously to changes in consumer spending triggered by changes in personal taxation. As Zidar (2019) demonstrates, reductions in state personal income taxes increase personal wages and employment, thereby leading to higher personal disposable income. Therefore, individual income tax cuts are likely to lead to higher consumer spending and, consequently, to higher firm investment opportunities and borrowing. To account for these possibilities, we also include event study indicators associated with state individual income tax cuts from $t = -2$ through $t \geq 4$ for all of our specifications.

4. RESULTS

4.1. Corporate tax cuts and corporate leverage

For our first tests, we use the regression specification in equation (1) to examine the relationship between corporate leverage and state corporate income tax cuts. We use the sample of private firms from the Y14 data spanning 2011–2017 and data on all state corporate income tax cuts since 2001. While this specification does not include tax hikes, we control for the incidence of tax hikes throughout our specifications, and we examine tax hikes separately below.

For this analysis, we split the sample at the median value of lagged total assets (\$288 million) to define small and large firms. This sample split intends to capture the different financial

frictions that small firms face relative to large firms. While financial frictions are unobservable, [Hadlock and Pierce \(2010\)](#) shows firm size is one of the most informative predictors of external financial constraints. Anticipating, this sample split also provides an interesting contrast for our model to reconcile.

Figure 1 presents the effects of tax cuts on leverage outcomes in event time, with additional details in [Online Appendix Table B1](#). We examine two main leverage measures: total debt and long-term debt, each scaled by beginning-of-period total assets. The x -axis of each panel contains the time since the event of the tax cut enactment. The dots represent the coefficient point estimates, $\tilde{c}_k$, in (1). The outer bars are the 90% confidence interval, and the inner bars are the 95% confidence interval.

Several results are of interest. First, as seen in Panel A, for our sample of small firms, we find significant positive responses of long-term leverage to tax cuts in event years zero through 4+. Small firms' long-term debt-to-assets ratios rise by almost 4% in the enactment year and remain elevated relative to pre-event levels 4 or more years later. Although our sample period is short, this evidence is consistent with the notion that leverage revisions are permanent. To understand the economic significance, we note that the average tax cut during this period was 1.2%. Because private firms in the U.S. paid an average tax rate of 21% ([Zwick et al., 2016](#)), this change represents approximately a 6% reduction in the tax bill. The final coefficient of .021 in year four also implies a change in leverage of about 6%. This large response is likely a product of the low-interest-rate environment during our sample period. A small change in the tax bill implies a large change in marginal q and a large response of both investment and the debt needed to finance it.

Second, in Panel B, we find that when we measure leverage using total debt, we obtain a positive, significant coefficient at the event date, but insignificant coefficients thereafter. This result makes sense because total debt contains short-term loans and revolving lines of credit, which are used to fund shorter-term expenses such as working capital. Third, in Panels C and D, we present the equivalent results for our sample of large firms. In neither case do we find significant coefficients, and in neither case does this result stem from wider confidence intervals. Finally, we do not detect any significant pre-trends for either small or large firms.

The dynamics evident in our results challenge two important alternative interpretations. First, one could imagine that the effects upon enactment and prior to the effective dates of tax changes are simply the reactions of firms taking advantage of the interest deduction before the tax cut. However, this hypothesis is inconsistent with the persistence of the leverage responses. Second, it is possible that our results are a product of credit supply effects, with banks shifting lending to states in which their own revenues are taxed at a lower rate. However, this hypothesis is inconsistent with the effects we find before the effective dates, as banks would have no incentive to shift lending to states before the new tax code becomes relevant. In contrast, risky lending rates can immediately impound expectations of future firm default probabilities that are induced by tax cuts. These rate shifts then affect firm incentives to borrow. We provide more direct evidence of this hypothesis below.

In Figure 2, we repeat the analysis in Figure 1, but we use effective instead of enactment dates of legislation, as in [Heider and Ljungqvist \(2015\)](#), with details in [Online Appendix Table B2](#). We find noisy and often statistically insignificant estimates. For example, for long-term debt, in Panel A, we find a significant positive effect in the subsample of small firms only at event date two. In contrast, we find no significant effects in Panel B for long-term debt or in Panels C and D, which correspond to the event studies for our subsample of large firms. We conclude that firms respond to enactments rather than the effective dates of tax cuts.

Next, we examine how tax cuts and hikes affect public firm leverage. For this analysis, we use data from Compustat for two subsamples: the same sample period that we use for the Y-14

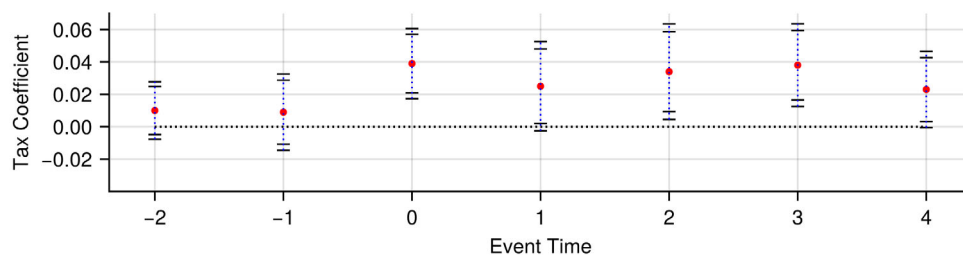
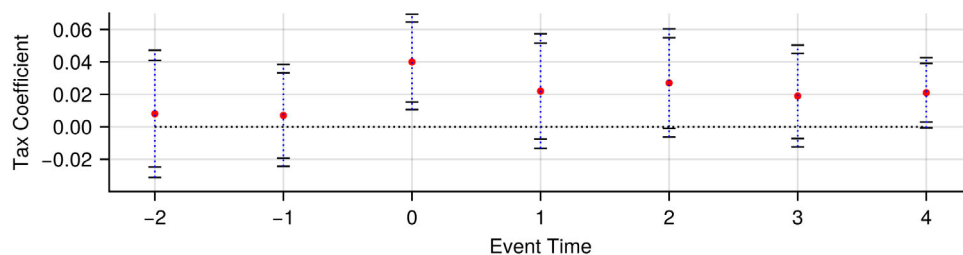
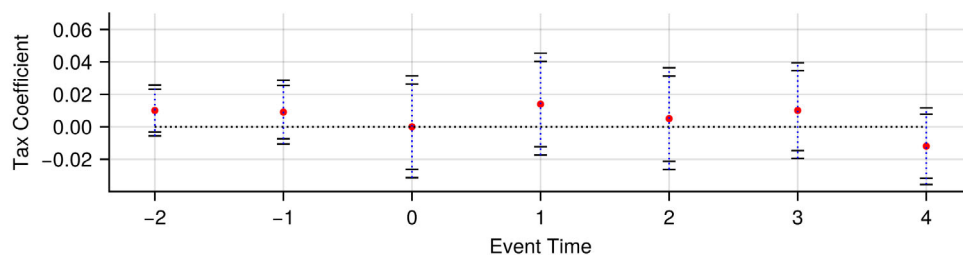
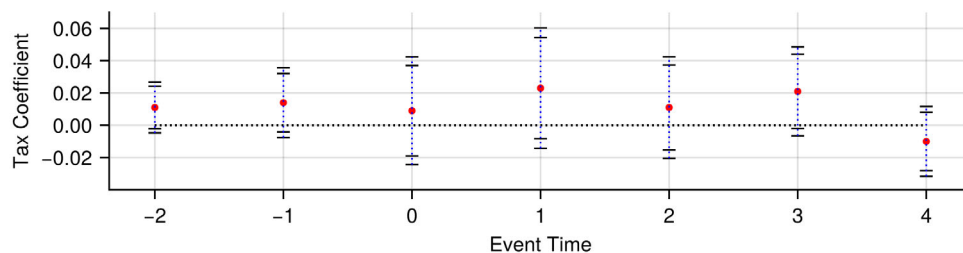
A Small Private Firms, Long-term Leverage**B** Small Private Firms, All Leverage**C** Large Private Firms, Long-term Leverage**D** Large Private Firms, All Leverage

FIGURE 1

Leverage event studies, enactment dates. This figure plots the coefficients from [Online Appendix Table B1](#), which contain estimates from firm-year panel regressions of the evolution of firm leverage around the enactment of state corporate income tax cuts. The data are our subsamples of small (Panels A and B) and large private firms (Panels C and D) from the Y-14 data. The dot represents the point estimate. The outer bars are the 90% confidence interval, and the inner bars are the 95% confidence interval

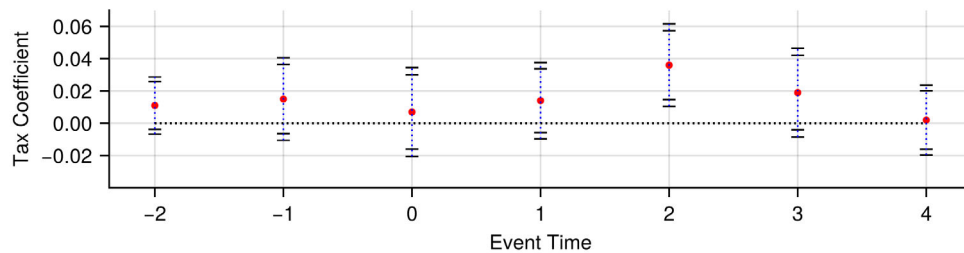
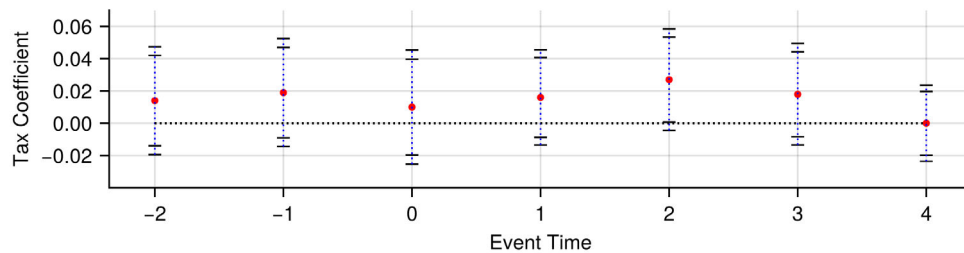
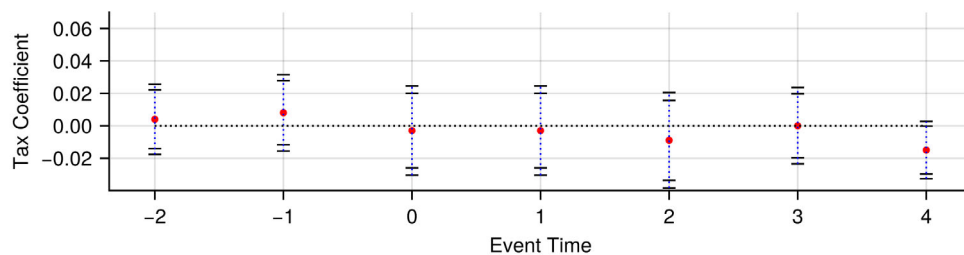
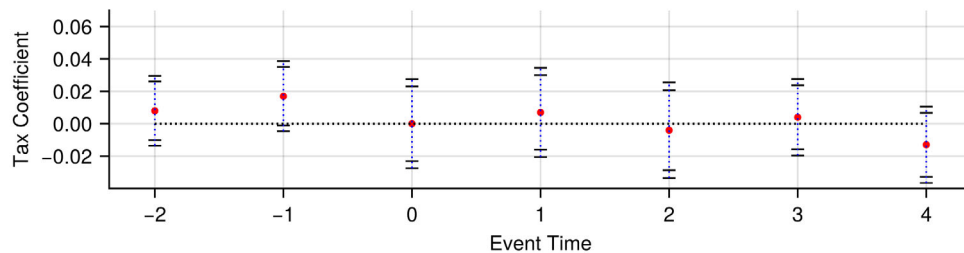
A Small Private Firms, Long-term Leverage**B** Small Private Firms, All Leverage**C** Large Private Firms, Long-term Leverage**D** Large Private Firms, All Leverage

FIGURE 2

Leverage event studies, effective dates. This figure plots the coefficients from [Online Appendix Table B2](#), which contain estimates from firm-year panel regressions of the evolution of firm leverage around the effective dates of state corporate income tax cuts. The data are our subsamples of small (Panels A and B) and large private firms (Panels C and D) from the Y-14 data. The dot represents the point estimate. The outer bars are the 90% confidence interval, and the inner bars are the 95% confidence interval

data, and a sample period that extends from 1989 to 2011, as in [Heider and Ljungqvist \(2015\)](#). Figure 3 shows the evolution of the book values of long-term debt scaled by total assets. We find no significant treatment effects for either the large or small firms in the early sample. For the late sample, we find a significant positive response in year two for tax cuts and several significant positive responses for tax hikes. However, in this latter case, we also observe significant pre-trends, so the results are difficult to interpret. Overall, evidence of an effect of taxes on leverage for public firms is scant.

Our results starkly contrast with the empirical relations documented in the prior literature. For example, [Titman and Wessels \(1988\)](#), [Heider and Ljungqvist \(2015\)](#), and [Faccio and Xu \(2015\)](#) all document that taxes are positively related to corporate leverage. A notable difference between these studies and ours is that we rely on samples of smaller privately held firms that face more financial frictions than public firms. In support of the notion that size and public status matter for the leverage-tax relation, our result of a negative relation is not without precedent, as [Farre-Mensa and Ljungqvist \(2015\)](#) also find a negative association between changes in book leverage and taxes in a sample of small private firms.

[Online Appendix Table B5](#) compares our methods to those in the most closely related study, [Heider and Ljungqvist \(2015\)](#) (HL, hereafter). Although we use the same sample period, because Compustat backfills, we cannot reproduce their exact coefficients. However, as seen in Column (1), our results closely approximate theirs. Like HL, we find no effect of tax cuts. For tax hikes, we find a significant estimate of 0.005, while theirs is 0.007. Next, we try to isolate the difference between their results and ours. In columns (3) and (4), we use their specification on our sample period, finding no effects. In columns (5) and (6), we return to their sample period, but we use enactment dates instead of effective dates. We find no effect of tax cuts but a significant negative effect of tax hikes. Next, the HL specification uses the change in long-term leverage as the dependent variable. We use the level of leverage because we have a dynamic specification. In Columns (7) and (8), we use a static specification with leverage as the dependent variable, and in Columns (9) and (10), we use our dynamic specification with leverage as the dependent variable. In both cases, we find no effects.

In sum, our results present two stylized facts to understand more deeply. First, the leverage of small private firms responds positively to enactments of tax cuts. Second, large private firms' leverage does not react to tax cuts.

4.2. *Real effects of tax cuts*

Tax cuts are also likely to have real effects via a direct positive effect on the marginal product of capital and a fall in the cost of capital resulting from changes in default thresholds, as argued in Section 1. In Figure 4, we examine the response of capital expenditures to the tax enactment dates, with the details in [Online Appendix Table B4](#). In Panels A and B, we consider investment scaled by tangible assets for small and large firms, and in Panels C and D, we similarly consider investment scaled by total assets. For both investment measures, for small firms, we find positive, significant responses between 0.015 and 0.021 at the enactment date. In both cases, we also find slightly smaller, but still significant responses in event year three. In contrast, we find no reaction of investment to the tax enactment dates for the large firms. These results are likely weaker than our leverage results because of time-to-build lags and investment lumpiness ([Doms and Dunne, 1998](#)), so changes materialize with some delay. Moreover, we cannot observe important components of investment, such as intangibles, in the Y-14 data, so we miss these potential effects. In sum, our results are consistent with the findings in [Zwick and Mahon \(2017\)](#) and [Ohrn \(2018\)](#) that the investment of small firms is more sensitive to tax incentives and that low-tax environments are associated with greater investment incentives.

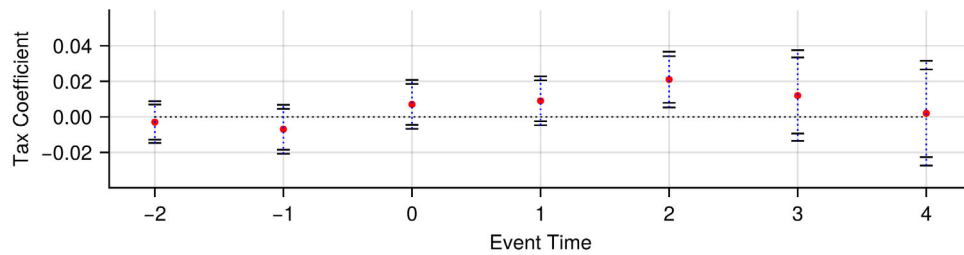
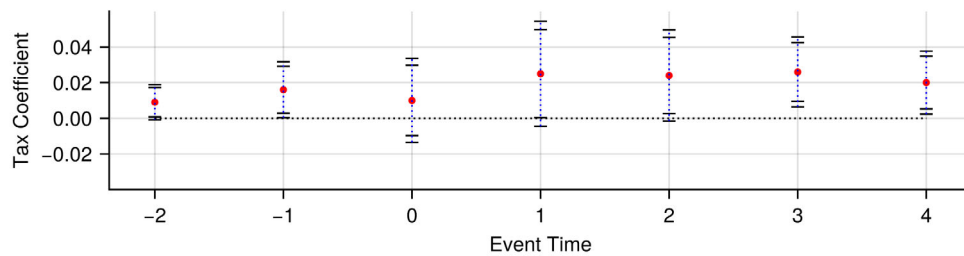
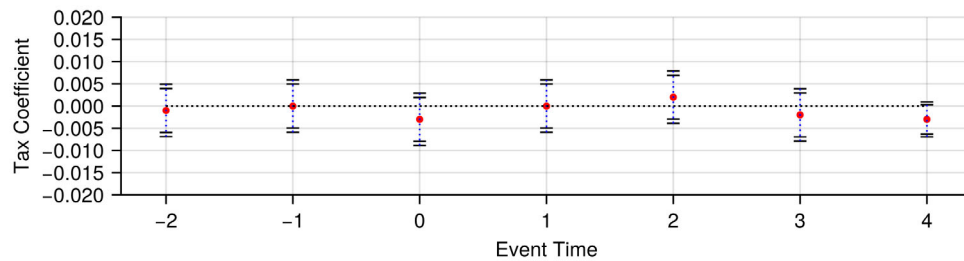
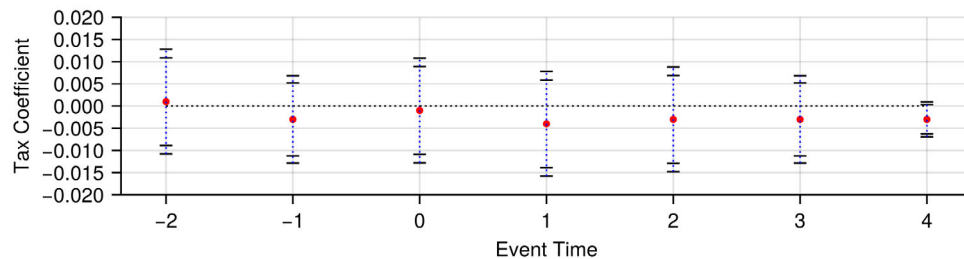
A Public Firms, Late Sample Period, Tax Cuts**B** Public Firms, Late Sample Period, Tax Hikes**C** Public Firms, Early Sample Period, Tax Cuts**D** Public Firms, Early Sample Period, Tax Hikes

FIGURE 3

Leverage event studies: public firms, enactment dates. This figure plots the coefficients from [Online Appendix Table B3](#), which contain estimates from firm-year panel regressions of the evolution of long-term firm leverage around the enactment dates of state corporate income tax changes. The sample runs from 1989 through 2017 and is restricted to public companies from the CRSP-Compustat database. Each regression contains both tax-hike and tax-cut dummies, and each pair of panels (A and B or C and D) corresponds to a single regression. Each dot represents a point estimate.

The outer bars are the 90% confidence interval, and the inner bars are the 95% confidence interval

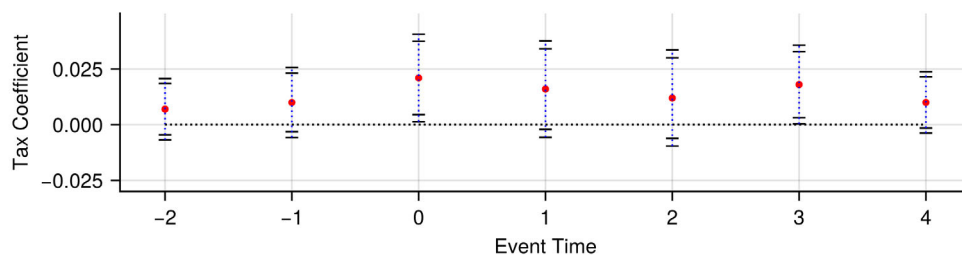
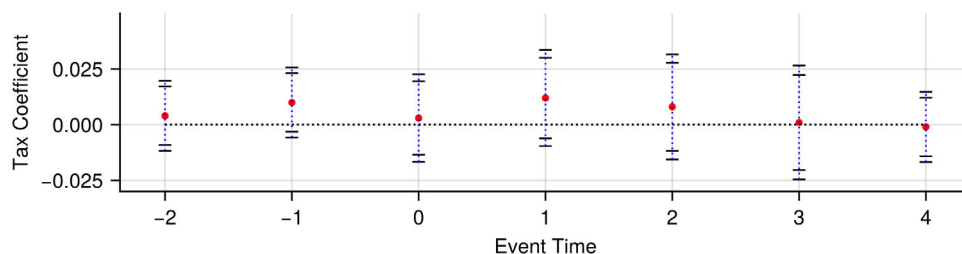
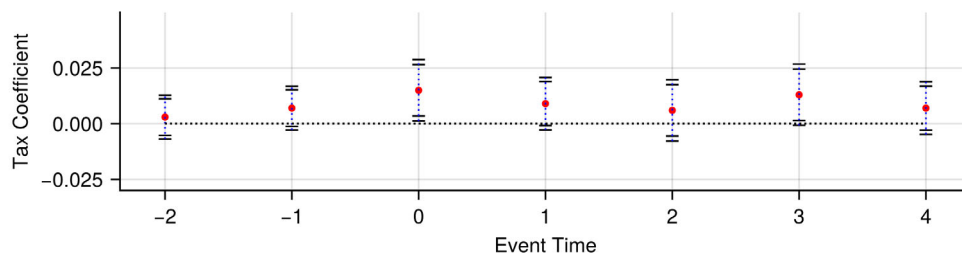
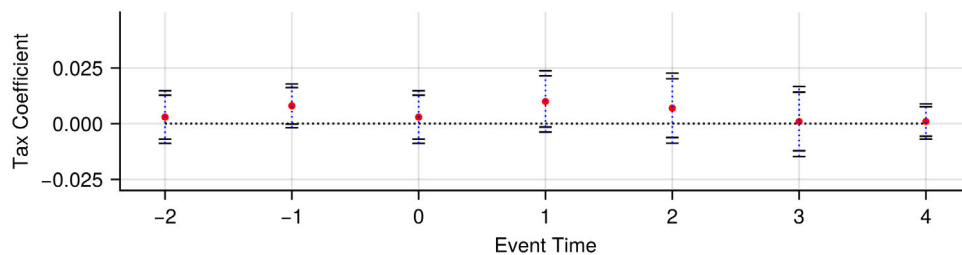
A Small Private Firms, Investment Scaled by Tangible Assets**B** Large Private Firms, Investment Scaled by Tangible Assets**C** Small Private Firms, Investment Scaled by Total Assets**D** Large Private Firms, Investment Scaled by Total Assets

FIGURE 4

Investment event studies, enactment dates. This figure plots the coefficients from Table B4, which contain estimates from firm-year panel regressions of the evolution of firm investment around the enactment of state corporate income tax cuts. The data are our subsamples of small (Panels A and C) and large private firms (Panels B and D) from the Y-14 data. Each dot represents a point estimate. The outer bars are the 90% confidence interval, and the inner bars are the 95% confidence interval

This evidence of real effects is important. We argue below that models without endogenous investment, such as those based on the framework in [Fischer *et al.* \(1989\)](#), cannot generate a negative relation between taxes and leverage, so evidence that taxes affect real outcomes lends credence to our choice of a theoretical framework below in which factor demand is endogenous.

4.3. *Alternative specifications*

Next, we examine the robustness of our results to several concerns, including heterogeneous treatment effects, equity issuance, the political economy of tax legislation, our thresholds for defining C corporations or small firms, and several additional specification issues.

As states enact tax legislation at different points in time, and as the size of tax changes differ across states, we investigate the robustness of our results to variation in treatment timing. [Borusyak *et al.* \(2024\)](#) show that in settings such as ours, treatment effect heterogeneity can bias event study estimates, as overall treatment effects are weighted averages of treatment-specific effects, with possibly negative weights. They propose an estimator that imputes the fixed effects estimated from untreated observations onto treated observations. We use this technique to reestimate our main specification, which uses the sample of small private firms, enactment dates, and long-term debt. In [Online Appendix Table B6](#), for the small firms, for event years 0–3, we find significant estimates that are similar to our baseline results in [Figure 1](#). For the large firms, we again find small, insignificant coefficients.

We also consider an estimator from [de Chaisemartin and D’Haultfoeuille \(2020, 2021\)](#) that is not affected by treatment effect heterogeneity and is based on variation from firms that switch to treatment (tax enactment) for the first time. In [Online Appendix Table B7](#), we find similar effects on leverage of .036 for small firms and no effect for large firms in the tax cut enactment year. We do not find any dynamic effects after the enactment year because the technique requires a complete panel for switchers in all 3 years following tax enactment, and we have a low number of remaining first-time switchers.

An important issue in our analysis is the endogeneity of the tax cuts, which can occur when politicians respond to changes in the economy with tax policy. While we do not find evidence of anticipation effects or pre-trends around tax enactments, these political economy concerns would nonetheless undermine the parallel trends assumption we make in our difference-in-differences analysis. Intuitively, if a tax cut happens during an economic boom when the government surplus is large, observed leverage changes might be due to the economic boom. To alleviate these concerns, we use the narrative approach from [Romer and Romer \(2010\)](#) and [Giroud and Rauh \(2019\)](#), which argues that tax changes are less likely to be endogenously related to economic activity if the primary objective of the tax legislation is to achieve a long-run goal or to deal with an inherited budget deficit or a surplus. Using this classification, we find that many of the tax changes in our sample can be classified as exogenous, with the only exceptions being Connecticut in 2011 and 2015, New York in 2014, and Rhode Island in 2014. [Online Appendix Table B8](#) shows that our leverage estimates are slightly larger when we exclude these states. We conclude that the leverage effects in [Figure 1](#) are unlikely to be driven by political economy concerns.

Next, we examine more closely whether the leverage changes in [Figure 1](#) result from equity adjustments instead of borrowing. This issue is important because our model is one of debt adjustments, so if our results are affected by equity adjustments, then a modification of our theoretical framework is in order. We consider this possibility unlikely for two reasons. First, since 2009, we observe firms raising finance through private equity markets in only 2,003 out of the 39,363 firm-year observations. Second, we redo the analysis for small private firms and long-term debt in [Figure 1](#), except we exclude firm-year observations belonging to firms that

have ever had access to private placement markets. In Columns (1)–(4) of [Online Appendix Table B10](#), we find similar event study estimates.

A further concern is that distressed companies might not have taxable profits, so our results might be an artefact of the tax advantage of debt not being relevant for a portion of our sample. We redo our analysis after excluding companies with bank internal ratings of “CCC” and lower. As shown in Columns (5)–(8) of [Online Appendix Table B10](#), our results are once again similar when we impose this restriction.

One limitation of our analysis is our inability to identify the tax status of individual firms. We address this issue in two ways. First, we change the C-corporation thresholds for inclusion in the sample from \$100 million in prior-year assets to \$75 or \$50 million. Second, we use information from the Statistics of U.S. Businesses (SUSB) Annual Data Tables, which show the share of companies organized as C corporations, S corporations, sole proprietorships, and LLCs by NAICS industry and size bucket. We then exclude the five NAICS two-digit industries in which the SUSB data indicate that pass-throughs (S corporations, LLCs, and partnerships) constitute significantly more than 50% of the firms with at least 500 employees: 61, 62, 71, 72, and 81. As we show in [Online Appendix Table B10](#), lowering the cutoff for C corporations results in tax coefficients of the same sign as those in [Figure 1](#), which contains our baseline results. However, both the size and significance fall. This result makes sense, as lowering the size cutoff introduces more pass-through entities into the sample. In columns (5)–(8), we show that removing the low-C-corporation industries makes our baseline coefficients in [Figure 1](#) slightly larger.

Next, the results in [Online Appendix Tables B11–B15](#), we consider several further specification choices. First, using the sample median to define small firms is arbitrary, so we use alternative size thresholds for small firms of \$150, \$200, \$250, or \$300 million in assets. Second, we include additional control variables typically used in leverage regressions, such as profitability, tangibility, and sales growth. Third, we consider the case of no controls. Fourth, we add dummy variables for corporate-tax and sales-tax effective dates to our specifications. Fifth, we consider two alternative measures of leverage: the log of debt and debt scaled by end-of-period assets, instead of current-period assets. Sixth, we include Census-region by year fixed effects to control for regional trends that might be correlated with tax cuts. In all but one case, we find similar results. The exception is the sample of small firms defined by the \$150 million cutoff, for which a small sample size means we find insignificant results.

Next, we consider the issue that U.S. corporations pay state corporate taxes to all states with which they have “nexus,” so the amount they owe to each state depends on apportionment formulas and the location of employment, property, and sales across states. Because we cannot merge the confidential Y-14 data with the administrative data necessary to construct nexus, we consider isolating those firms that operate in a single state. We take two approaches. First, if either the NETS or the S&P Cross Reference Database identifies a company as multi-state in any year, we exclude it from the sample. Second, we exclude firms that borrow through subsidiaries located in multiple states during the entire sample period, according to the Y-14 data. [Online Appendix Table B16](#) shows that these sample restrictions produce results that are similar to our baseline findings in [Figure 1](#), with positive responses of leverage for small firms and with no evidence of a response in large firms.

One issue with our event studies is the use of dummy variables for tax changes. This choice implies that small tax changes influence our results as much as large changes and possibly drive our results. To investigate this possibility, in columns (1)–(4) of [Online Appendix Table B17](#), we present results when we limit the tax cuts that are at least 0.5% and at least 1%. Not surprisingly, when we consider these smaller sets of more significant events, we find slightly stronger effects with patterns of significance that are similar to those in our baseline results in [Figure 1](#).

Next, we consider tax hikes. In Columns (5)–(8) of [Online Appendix Table B17](#), we repeat our baseline specification for the large and small Y-14 firms. In column (5), we find that hikes result in a fall in leverage for small firms in event year 4. In column (6), we also find falls in leverage in event years 1, 2, and 4 for large firms. We find similar results in columns (7) and (8) when we omit a tax hike used only to define a pre-trend and states with tax hikes not characterized as exogenous, according to the methodology in [Giroud and Rauh \(2019\)](#). While these results confirm our general intuition, they are less credible than our baseline results in [Figure 1](#) because we do not have enough tax hike observations to identify pre-trends.

This result prompts the question of whether the effects of tax cuts and hikes are symmetric. Because the Y-14 sample period is short, we turn to the data on firms' syndicated bank borrowing from the Shared National Credit Program, which spans 1992 through 2018. Our dependent variable is the log of total syndicated commitments to a given firm in a given year. Because there are no balance-sheet variables in the SNC data, we cannot scale commitments by total assets, so we use the log. [Online Appendix Table B18](#) indicates that total firm commitments increase around corporate tax cuts, as in our main specifications. However, these results are slightly smaller in magnitude, and with one exception, of marginal significance. In contrast, we find large and significant reductions in borrowing following tax hike enactments, with the average firm lowering commitments by 6–7%. Because the relative magnitudes of hikes and cuts over this period are roughly the same, we conclude that tax hikes have a larger effect. This result makes intuitive sense, as reducing debt incurs few, if any, costs, while increasing debt often entails costs. However, these results are also less robust to specification choices because we cannot observe firm size or firm-level controls.

5. MODEL

To understand the underlying economics behind the empirical patterns in our data, we turn to a dynamic model of an equilibrium economy. We first establish a baseline model, estimate its parameters, and analyze the optimal policies. We finish by exploring several different assumptions about the stringency and types of financial frictions in the model.

The economy consists of a representative consumer and a unit continuum of firms. The economy also contains a government and a financial intermediary sector, but these entities simply act as pass-through agents for the firms and the consumer. There are no aggregate shocks to consumers or firms in the economy.

Each of the infinitely lived firms uses capital and labour in a stochastic, decreasing returns-to-scale technology to generate output, y , according to

$$y = z^\nu (k^\alpha n^{1-\alpha})^\theta, \quad (3)$$

where k is the stock of capital, n is labour, z is a productivity shock, α is capital's share, and θ governs the degree of returns to scale. We normalize the parameter ν to be $1 - (1 - \alpha)\theta$. In addition to this basic technology, we assume that the firm has a fixed component of operating costs, which we denote as f . The productivity shock, z , is lognormally distributed and follows a process given by:

$$\ln(z') = \rho \ln(z) + \sigma_z \varepsilon', \quad \varepsilon' \sim \mathcal{N}(0, 1), \quad (4)$$

where a prime indicates the subsequent period, and no prime indicates the current period.

Investment in capital, I , is defined by a standard capital stock accounting identity:

$$k' \equiv (1 - \delta)k + I, \quad (5)$$

in which δ is the rate of capital depreciation. The price of the capital good has been normalized to one. Adjusting the capital stock incurs quadratic costs that take the form:

$$AC(k, k') = \frac{\psi(k' - (1 - \delta)k)^2}{2k} \quad (6)$$

where ψ is a parameter that governs the magnitude of adjustment costs.

Taxation in our model is simple, as there is only corporate taxation at a stochastic rate τ , which follows an autoregressive process given by:

$$\tau' = \mu(1 - \rho_\tau) + \rho_\tau \tau + \sigma_\tau u'. \quad (7)$$

where μ , ρ_τ , and σ_τ are the mean, serial correlation, and innovation standard deviation of the tax process. u' has a truncated normal distribution, which ensures the tax rate stays positive. This tax rate applies both to profits and financing activities, as described below.

The firm can finance its optimal investment program with external debt or internal equity in the form of retained earnings. We let p denote the stock of net debt, so $p > 0$ indicates that the firm has debt on the balance sheet, and $p < 0$ indicates that the firm has cash on the balance sheet. We assume that debt is raised through a competitive financial intermediary sector, which in turn raises the necessary funds from the representative consumer. Debt is a one-period discount bond on which the firm can default. Let the interest rate on debt be $\tilde{r}(k', p', z, \tau)$, so debt proceeds are $p'/(1 + r(k', p', z, \tau)(1 - \tau))$.² As outlined below, this interest rate is determined endogenously from the lender's zero-profit condition and is consequently a function of the model's state variables. If the firm opts to save instead, it earns the after-tax risk-free rate, r , with the interest taxed at a rate τ . Thus, the interest rate on debt can be expressed as:

$$r(k', p', z, \tau) = \begin{cases} \tilde{r}(k', p', z, \tau) & \text{if } p > 0 \\ r & \text{if } p \leq 0 \end{cases} \quad (8)$$

Cash flows to shareholders, $e(k, p, n, k', p', z, \tau)$, are then the firm's after-tax operating income plus net debt issuance, minus net expenditure on investment, and minus tax-deductible interest payments on debt, as follows:

$$e(k, p, n, k', p', z, \tau) = (1 - \tau) \left(z^\nu (k^\alpha n^{1-\alpha})^\theta - wn - f \right) - (k' - (1 - \delta)k) - AC(k, k') + \frac{p'}{1 + r(k', p', z, \tau)(1 - \tau)} - p, \quad (9)$$

where w is the wage rate, which is determined in equilibrium.

While a positive firm cash flow is distributed to the consumer, we assume that negative cash flows are not allowed, that is:

$$e(k, p, n, k', p', z, \tau) \geq 0. \quad (10)$$

2. This formulation assumes that the firm takes the tax deduction when it issues the debt. While not in accord with real-world debt contracts in which the tax advantage accrues only later when interest is paid, this assumption reduces the state space and simplifies the default condition (Strebulaeu and Whited, 2012).

This assumption eliminates external equity finance. Because our sample firms are private, they have no access to public equity markets and limited access to private equity markets, so this assumption reflects the tailoring of the model to our sample. Regardless, we relax the assumption below when we examine the robustness of the model.

The Bellman equation for the problem can then be expressed as:

$$\Pi(k, p, z, \tau) = \max_{k', n, p'} \left\{ e(k, p, n, k', p', z, \tau) + \frac{1}{1+r} \mathbb{E} \Pi(k', p', z', \tau') \right\}, \quad (11)$$

subject to (5) and (10).

5.1. Loan contract

We assume that a perfectly competitive financial intermediary sector offers the firm a one-period loan contract that need not be fully collateralized, so the firm can default. We follow [Gilchrist et al. \(2013\)](#) and [Michaels et al. \(2019\)](#) by assuming that the firm's future market value is not collateralizable, so it defaults if it does not have enough resources on hand to repay its debt. While this assumption is particularly apt for our sample of smaller private firms, we examine below how the model behaves when we adopt a more lenient default condition in which lenders can extend credit as long as the firm has a positive equity value ([Hennessy and Whited, 2007](#); [Gao et al., 2021](#)).

Default is triggered when debt repayment exceeds the firm's current after-tax profit plus the fraction, $1 - \xi$ of its capital that can be recovered in default:

$$(1 - \tau) \left(z^\nu (k^\alpha n^{1-\alpha})^\theta - wn - f \right) + (1 - \xi)(1 - \delta)k \equiv R(z, \tau, k, p) < p. \quad (12)$$

We subtract the wage bill from output in (12) because labour is paid in full, even if the firm defaults. Also, taxes get paid before the lender can recover any payments. Both of these assumptions follow absolute priority rules. Finally, because the tax deduction is taken when the firm issues debt, it is absent from this condition. For fixed levels of (k, n, p) , (12) defines a region, denoted by Ω , over the joint domain of (z, τ) in which default occurs.

Given this default threshold, the contractual interest rate, $\tilde{r}(k', p', z, \tau)$, is determined by a zero-profit condition that must hold under free entry in the intermediation sector. The payoff to the lender outside of default is simply this contractual interest rate. Inside default, the lender recovers an amount equal to the left side of equation (12). Thus, under free entry and risk-neutrality, the face value of debt discounted at the risky rate, $\tilde{r}(k', p', z, \tau)$, must equal the expected payoff discounted at the risk-free rate. Therefore, $\tilde{r}(k', p', z, \tau)$ satisfies:

$$\frac{p'}{1 + \tilde{r}(k', p', z, \tau)} = \frac{1}{1+r} \left[\int_{\Omega} (R(z', \tau', k', p')) dG(z', \tau' | z, \tau) + (1 - G(\Omega | z, \tau)) p' \right]. \quad (13)$$

For a given (p', k', z, τ) , equations (12) and (13) pin down the loan contract.

5.2. Equilibrium

The economy also contains an infinitely-lived representative consumer who chooses consumption and labour each period to maximize the expected present value of her utility, with a discount factor of b . Her one-period utility function is given by $\ln(c) + \varphi(1 - n_s)$, in which c is consumption, n_s is the supply of labour, and φ is a parameter that governs the utility of leisure.

She has three sources of income: labour, dividends, and the net tax revenue generated from the firms, T , which the government transfers to the consumer as a lump sum. Her one-period budget constraint is given by:

$$c + p'_d - p_d(1 + r) = wn_s + \bar{e} + T, \quad (14)$$

in which p_d is consumer wealth, and $\bar{e}$ is aggregate firm dividends.

Let ζ be the stationary distribution over the firm's states, (z, τ, k, p) . We define equilibrium in this economy as follows.

Definition 1. A competitive equilibrium consists of (i) optimal firm policies for capital, labour, and debt, $\{k', n, p'\}$, (ii) allocations to the consumer of consumption, c , and labour, n_s , and (iii) prices, (w, r) , such that:

1. All firms solve the problem given by (11).
2. The consumer maximizes her utility, subject to (14).
3. The labour, bond, and output markets clear.

$$n_s = \int n d\zeta \quad (15)$$

$$p_d = \int p' d\zeta \quad (16)$$

$$c = \int (y - I - AC) d\zeta. \quad (17)$$

5.3. Solution

We solve the model using policy-function iteration and bisection, which yields an equilibrium wage and interest rates, a value function, $\Pi(k, p, z, \tau)$, and policy functions for capital and debt, given by $k'(k, p, z, \tau)$ and $p'(k, p, z, \tau)$. We simulate the model by shocking the policy functions, given the processes for z and τ in equations (4) and (7), and tracing out the paths of optimal policies. In all simulations, when a firm defaults, it is replaced with a firm that, at the default date, lies just below the default threshold but that then receives a shock vector (z, τ) from the distribution of (z, τ) that makes it viable.

5.4. Estimation

We estimate the model parameters using our sample of small firms from the Y14 data. We set a subset of our parameters outside the model to simplify computation. First, we set the consumer's discount factor equal to 0.96. Second, we set $\varphi = 2.5$, which implies that the consumer spends approximately one-third of her time working. Third, we set the average tax rate, μ in (7) equal to 0.2, following Nikolov and Whited (2014). This rate is lower than the statutory national rate because it accounts for personal taxes, as in Graham (1996). Finally, we set the capital share parameter equal to 0.3.

Next, we estimate the standard deviation and serial correlation of the tax process (7) using a low-dimensional simulated method of moments procedure. We simulate data from (7) and then match the following two targets in the simulated and real data: the average absolute value of the tax changes and the average length of the spell between tax changes. Because the estimation of this last quantity can be affected by truncation bias, we ensure that this source of bias is the same

in the simulated and real data by simulating the tax process in equation (7) 10,000 times, with each simulation having the length of our sample period. We estimate ρ_τ to be 0.662 and σ_τ to be 0.022.

We estimate the remaining parameters (θ , σ , ρ , δ , ψ , ξ , and f) jointly by minimizing the distance between a list of targets constructed from model-simulated data and those computed with actual data. In this estimation procedure, we use the optimal weight matrix, as in [Bazdresch et al. \(2018\)](#), clustered by firm and year. [Online Appendix B](#) provides variable definitions for our actual data. In our simulated data, leverage, operating profits, and investment are given by p/k , $(z^v(k^\alpha n^{1-\alpha})^\theta - wn - f)/k$, and $(k' - (1 - \delta)k)/k$.

We choose the following nine moments to match. The first six are the means and standard deviations of debt, investment, and operating income, all expressed as a ratio of assets. We also include the serial correlation of investment and operating income, which we calculate using the method in [Han and Phillips \(2010\)](#) to account for firm fixed effects. The last moment is an estimate of the coefficient in a two-way fixed effects regression of leverage on tax-change event dummies, with a cut coded as 1 and a hike coded as -1 , to correspond to our empirical specifications, so a positive coefficient implies that a cut boosts leverage.

We discuss both global and local identification. For the former, we consider the sensitivity of moments to model parameters. As such, this exercise informs the behaviour of the model but not the data. While all of the model parameters affect all of the moments, some of these moments are particularly useful for parameter identification. In [Online Appendix Figure B2](#), we plot the relations in the model between selected moments and parameters, with the baseline parameters from the small-firm estimates in [Table 2](#). First, the mean and standard deviation of investment help identify the capital depreciation rate, δ , and the adjustment cost parameter, ψ , respectively. In this class of models, steady-state investment rises with the depreciation rate, and the variance of investment naturally declines as quadratic adjustment costs induce more smoothing. Accordingly, the serial correlation of investment also helps identify ψ . The serial correlation, ρ , of the process for z is directly related to the estimated serial correlation of operating income, and all model variances increase with the standard deviation, σ , of the driving process, with the effect on the variance of operating income being the strongest. Mean operating income is mechanically decreasing in the profit function curvature, θ , and in the fixed cost of production, f . We can separately identify these two parameters because the variance of operating profits is also mechanically decreasing in θ . In contrast, this variance is almost invariant to the fixed cost of production. Finally, leverage is useful for identifying the ξ , as higher values of ξ lead to lower recovery in default and thus lower leverage.

We also consider a local measure of identification from [Andrews et al. \(2017\)](#). This diagnostic reflects both the precision of the data moments and local versions of model-based moment-parameter sensitivities. However, it often does not reflect the underlying economics of the model because the diagnostic reflects nonlinear combinations of model sensitivities and data covariances. In [Online Appendix Figure B3](#), we present a heat map representing this diagnostic. Each cell can be interpreted as the effect of a one standard deviation change in a moment on a parameter. The color coding ranges from dark purple (-0.02) to dark green (0.02). While most parameters are influenced by several moments, the depreciation rate, δ , is influenced only by the means of profits and investment.

[Table 2](#), Panel A reports the model parameter estimates for our samples of small and large firms. All but one of the parameter estimates are significantly different from zero, with the exception being the adjustment cost parameter for the small firms. The estimates of the profit function curvature, depreciation rate, and the standard deviation and serial correlation of the driving process are within ranges typically reported for this class of models ([Bazdresch et al., 2018](#)). The fixed operating costs, f , for both estimations are around 0.15, amounting to just under half of

TABLE 2
Model estimation

A. Parameter estimates							
	θ	ρ	σ	δ	ψ	ξ	f
Small Firms	0.9268 (0.0328)	0.6305 (0.1953)	0.0992 (0.0468)	0.0865 (0.0021)	0.1106 (0.1219)	0.6009 (0.0402)	0.1641 (0.0598)
Large Firms	0.9001 (0.0173)	0.6215 (0.0915)	0.1506 (0.0271)	0.0858 (0.0019)	0.9955 (0.3412)	0.3218 (0.1563)	0.1536 (0.0318)

B. Targeted moments							
	Small firms			Large firms			
	Actual	Simulated	<i>t</i> -stat.	Actual	Simulated	<i>t</i> -stat.	
Mean debt-to-assets	0.3677	0.3697	−0.2150	0.3944	0.3786	1.7983	
Standard deviation debt-to-assets	0.1066	0.1137	−6.6722	0.1131	0.0738	0.9987	
Mean investment-to-assets	0.1439	0.0917	0.4649	0.1131	0.0866	0.3901	
Standard deviation investment-to-assets	0.0817	0.0984	−0.3893	0.0742	0.0424	0.2091	
Serial correlation investment-to-assets	0.5925	0.2870	0.1320	0.5635	0.4288	0.0873	
Mean income-to-assets	0.1547	0.1528	0.0399	0.1351	0.1731	−0.9351	
Standard deviation income-to-assets	0.0593	0.0314	0.0690	0.0546	0.0531	0.0214	
Serial correlation income-to-assets	0.8483	0.7190	0.1855	0.7908	0.6116	0.0438	
Tax elasticity	0.0214	0.0173	0.0019	−0.0062	−0.0023	−0.0027	

Notes: This table reports the parameters and moments from the estimation of our model from Section 5. We use the subsamples of small and large firms from the Y-14 data. The sample period is 2011–2017. The parameter θ is the curvature of the production function; ρ is the serial correlation of the shock process; σ is the standard deviation of the shock process; δ is the capital depreciation rate; ψ governs quadratic capital adjustment costs; ξ is the deadweight cost of default; and f is the fixed operating cost. “Debt-to-assets” is defined as long-term debt divided by lagged total assets; “investment-to-assets” is defined as net capital expenditures divided by fixed assets; “income-to-assets” is defined as EBITDA divided by lagged total assets; “debt issuance” is defined as the change in debt scaled by lagged total assets. The *t*-statistics next to the simulated and actual moments are based on the null hypothesis of the equality of each of these moment pairs.

steady-state operating profits. Overall, the parameter estimates for the large and small firms are quite similar, with two important exceptions. The estimates of the deadweight loss in default are 0.32 and 0.6 for the large and small firms, respectively, indicating elevated financial frictions for the small firms. Although seemingly high, these estimates are in line with the average recovery rate found in [Kermani and Ma \(2023\)](#). In addition, the estimate of the adjustment cost parameter for the large firms is nine times larger than the estimate for the small firms. We attribute this difference to large firms comprising an aggregation of asynchronous projects, so their investment is smooth, as in [Bloom *et al.* \(2007\)](#).

Table 2, panel B reports the model and data moments. Importantly, we match the coefficient in a two-way fixed-effects regression of leverage on a tax change dummy. For neither the large nor the small firms is the moment difference significantly different from zero. Moreover, the model matches both the sign and economic magnitude of these coefficients. This result is important because it shows that the model can generate either a positive or negative leverage response to a tax cut, depending on the parameter values. Although with one exception, the simulated and data moments are insignificantly different from each other, one economically but not statistically significant discrepancy occurs for the standard deviation of investment in the sample of large firms. This pattern occurs because this moment is imprecisely estimated.

The model also roughly matches the coefficient from regressing debt issuance scaled by assets on leverage, an untargted moment that characterizes leverage dynamics. For small firms,

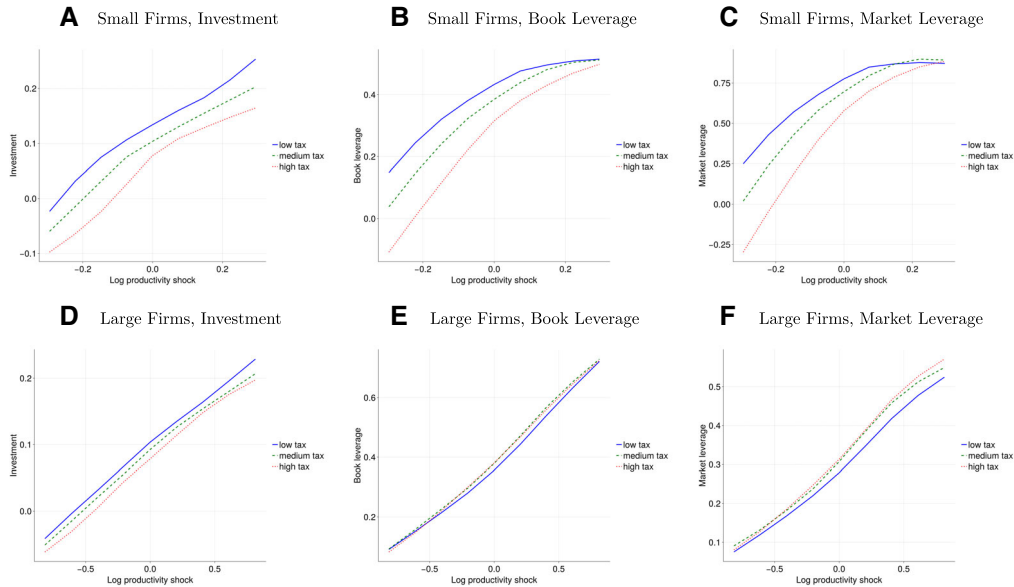


FIGURE 5

Optimal Policies and Taxes. This figure plots three policies as a function of the profitability shock: investment (I/k), the ratio of debt to capital (p/k or Book leverage), and the ratio of debt to the sum of debt and the market value of equity ($p/(v+p)$ or Market leverage). Each line represents a different level of the tax rate: low, medium, and high. The plotted policy functions are averages over the ergodic distribution of the model, conditioning upon the indicated values of the profitability shocks. Panels A–C depicts optimal policies for parameters estimated on our sample of small private firms, and Panels D–F depicts optimal policies for a different set of parameters estimated on our sample of large private firms

the coefficient in the model is -0.26 , while in the data, it is -0.65 . For large firms, the model and data coefficients are -0.09 and -0.47 , respectively. These coefficients approximate persistence, as they are roughly one minus a coefficient in an $AR(1)$ model. Thus, while in both cases, the model sensitivity is smaller, it captures the modest leverage persistence seen in the data.

5.5. Optimal policies

Given these two parameterizations, we now explore the aspects of the model that produce different sensitivities of leverage to taxes. First, in Figure 5, we show the behaviour of optimal investment, I/k , book leverage, p'/k' , and market leverage, V'/k' . Panels A–C correspond to the model parameterization from the small private firms, and Panels D–F correspond to the large private firms. Below, we refer to these parameterizations using the terms “small firms” and “large firms.” Each plot depicts one of these policies as a function of the productivity shock, z , and each line corresponds to a different tax rate (0.12, 0.20, 0.29). In all cases, the optimal policies are averaged over current leverage, p/k , and current capital, k , using the model steady-state distribution, ζ .

First, comparing Panels A and D, we see that small-firm investment responds more to the z than large-firm investment. If the log shock moves from -0.25 to 0 , investment rises by approximately .11 for the small firms at the median tax rate and .05 for the large firms. This response is a product of the much larger estimate of capital adjustment costs for large firms.

In Panels B and E, for both the large and small firms, optimal book leverage rises with the productivity shock, consistent with the evidence of procyclical debt issuance from [Covas and Den Haan \(2011\)](#). As capital productivity increases, firms find it optimal to transfer resources through time via capital rather than a storage technology, which in our model is $-p$. Equivalently, when productivity rises, optimal investment outlays outpace internal resources, and the firm opts for debt finance. While leverage policy rises with z for both model parameterizations, for the large firms, a higher tax rate produces slightly higher optimal debt (Panel E), while for the small firms, a higher tax rate optimally results in markedly lower leverage (Panel B). This result in the model is consistent with our own empirical evidence that the negative relation between debt and taxes is negative for small firms and slightly positive for large firms.

Next, comparing Panels B and C, we find that for the small firms, the responses of market and book leverage to the tax rate and the z shock are nearly identical. As seen in Panels E and F, for the large firms, the responses of market and book leverage to the tax rate are similar, with high tax rates implying higher leverage. However, for the large firms, market leverage falls with z . When z rises, capital and leverage respond weakly, but value rises more sharply, so the ratio of debt to the market value of the firm falls.

Part of the intuition behind the difference between the small and large firms in Figure 5 stems from the upward slope of the book-leverage policy functions. A fall in the τ shock makes capital more attractive relative to a storage technology ($-p$), just as a rise in the z shock does. For the model parameterization from the sample of small firms, this effect outweighs the increase in the tax benefit because these firms face small capital adjustment costs. For the model parameterization from the sample of large firms, this effect does not outweigh the tax advantage, as there is little substitution between capital and a storage technology in the face of high capital adjustment costs.

Further intuition behind the difference between the small and large firms in Figure 5 can be found in Figure 6. Panels A and C depict, for the small and large firms, respectively, the endogenous after-tax risky interest rate on debt, $\tilde{r}(k', p', z, \tau)$, as a function of leverage for the median size firm, with each line corresponding to a different tax rate. The effect of leverage on the contract is intuitive, as higher inherited leverage in any period raises default risk and raises the risky rate.

In both cases, the effect of taxes depends on the level of leverage. When leverage lies below .47 for the small firms and .74 for the large firms, debt is safe, and the after-tax risky interest rate mechanically falls as the tax rate rises, reflecting the interest tax deduction. However, for higher levels of leverage, the relation between taxes and the after-tax risky interest rate reverses. When leverage is high, the after-tax interest rate rises monotonically in the tax rate, as a higher tax rate makes firms less profitable and more likely to default. This lower default threshold is, in turn, capitalized into the risky rate.

Panels B and D of Figure 6 show the relation for the median-leverage firm between the after-tax risky loan rate and the capital stock, for the small- and large-firm estimations, respectively. As seen in Panel B, for the small firms, the firm experiences higher equilibrium rates when it has low capital, but in Panel D, we find no such relation for the large firms. The high rates in Panel B occur because of the fixed cost of production. Low-capital firms are less likely to be able to cover the fixed cost and are more likely to default, so the default premium is higher.

More generally, Figure 6 shows that for the model based on the large firm estimates, the firm is unlikely to be in a range in which a higher tax rate results in a higher after-tax loan rate, while for the model parameterization based on the small firms, this event is possible and contributes to the negative effect of taxes on leverage. Quantitatively, equation (12) can be scaled by the capital stock to offer a measure of the distance to default, with a larger measure indicating a larger precautionary cushion from default. For the small and large firms, the average distances

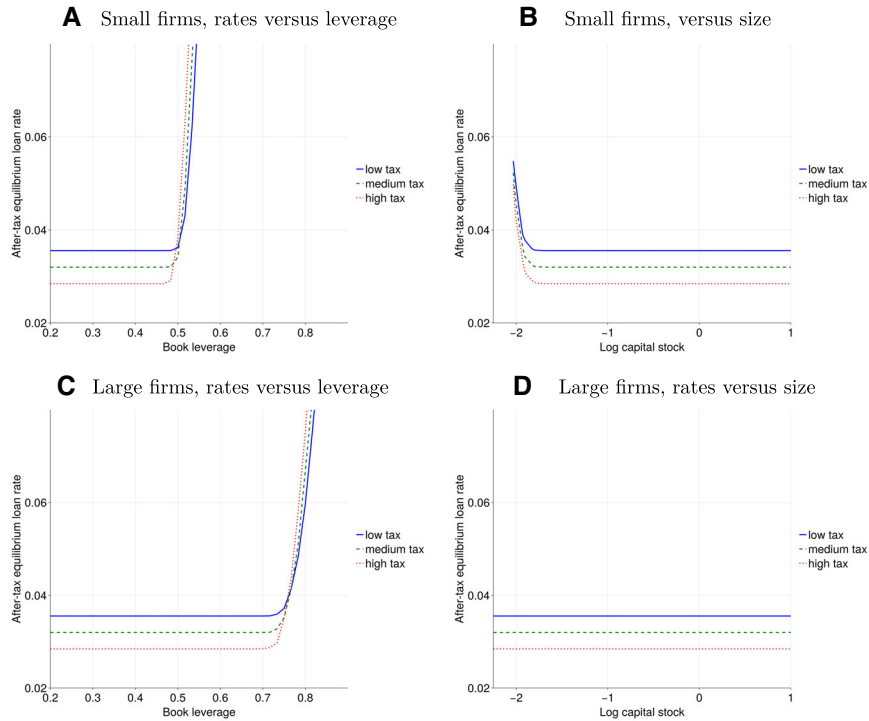


FIGURE 6

After-tax equilibrium loan rates. This figure plots after-tax, equilibrium risky loan rates as functions of firm size and firm leverage. Panels A and B depict loan rates for parameters estimated on our sample of small private firms, and Panels C and D depict loan rates for parameters estimated on our sample of large private firms

to default are 0.14 and 0.39, respectively. Intuitively, small firms find it optimal to locate nearer to the default threshold because of their optimally more aggressive investment policies that they need to finance. As such, they are more likely to be near the region in which higher taxes result in higher risky debt rates.

In sum, two features of our model offset the baked-in tax benefit, allowing leverage to respond negatively to taxes. First, lower corporate taxes induce looser default thresholds, which are capitalized into lending rates. Second, a lower tax rate raises the marginal product of capital, so firms find it optimal to substitute away from cash and into capital by using more leverage.

5.6. Comparative statics and counterfactuals

We examine how the sensitivity of leverage to a tax change responds when we change four parameters in the model: the volatility and serial correlation of the tax process in (7), the capital adjustment cost, and the deadweight cost of default. The results from this exercise are in Figure 7, where each panel is constructed by starting at the parameterization of the model from the sample of small firms. Then we let the parameter in question take on 10 values between the endpoints indicated on the x -axis of each panel, solve the model for each value, and plot the tax sensitivity from each solution. Recall that this sensitivity is computed so that a positive sign implies that a tax hike reduces leverage.

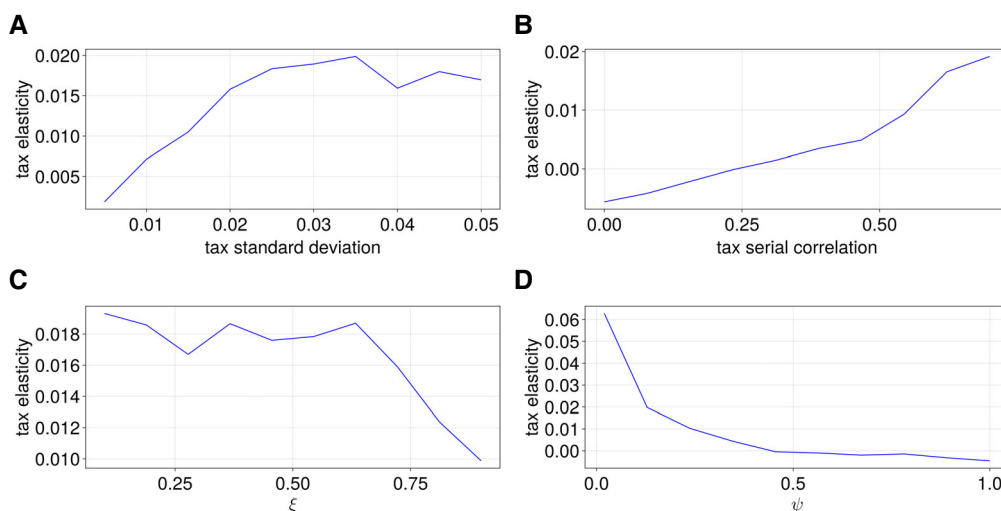


FIGURE 7

Determinants of Debt–Tax sensitivity. All panels depict the effects of different model parameters on the sensitivity of leverage to a dummy representing a tax change. In Panel A, the parameter is the volatility of the tax process. In Panel B, the parameter is the serial correlation of the tax process. In Panel C, the parameter is the deadweight cost of default. In Panel D, the parameter is the capital adjustment cost

In Panel A, we find that the volatility of the tax process has a strong positive effect on the tax sensitivity. This effect is mechanical, as higher volatility implies that the tax changes are larger, so a regression on a dummy variable produces a larger coefficient. In Panel B, we find that this sensitivity rises sharply with the serial correlation of the tax process, especially for high levels of serial correlation. Intuitively, firms react more strongly to taxes when they expect the changes to persist. In Panel C, we see a sharp negative relation between the tax sensitivity of leverage and the parameter that represents the deadweight costs of default. When these costs are low, firms use much more debt and are likely to be close to the region in Figure 6 in which the risky after-tax interest rate responds positively to the tax rate. In Panel D, we see a negative relation between the tax sensitivity of debt and capital adjustment costs. For very high adjustment costs, this sensitivity flips sign so that a cut implies lower leverage, as a standard textbook model would imply. This result underlies the intuition behind the wrong-way sensitivity of leverage to taxes in Figure 5. When adjustment costs are low, firms easily substitute between capital and a storage technology ($-p$), so this effect outweighs the tax benefit. When adjustment costs are high, this substitution is dampened substantially, and the effect of the tax benefit is stronger.

Next, we consider the misallocation and welfare implications of removing the tax benefit of debt altogether while leaving in place two frictions that also shape corporate leverage policy: profit taxation and no access to external equity finance.³ The outcome of this experiment is not immediately clear because we are comparing two second-best outcomes, both of which contain distortionary frictions. For each economy, we compute consumer welfare and the measure of

3. See [Glover *et al.* \(2015\)](#) for a related experiment that focuses on firms' default outcomes. We do not consider the 2017 Tax Cuts and Jobs Act, which capped interest deductibility at 30% of income instead of eliminating it. [Carrizosa *et al.* \(2023\)](#) show that only 15% of U.S. public firms were affected by this provision.

capital-labour misallocation from [Hsieh and Klenow \(2009\)](#), which captures the output loss from dispersion in marginal products across firms.

For the model parameterization based on the sample of small firms, when we remove the tax deduction, leverage falls from 0.38 to -0.45 . This significant fall is consistent with our model's overall negative effect of corporate taxation on leverage because we remove only the benefit of debt, leaving all costs intact. The misallocation measure rises by 2.1%, and welfare also drops. Without the tax benefit of debt, firms behave in a precautionary manner by hoarding cash, as they want to save their way out of bumping into the constraint in equation (10). This constraint additionally implies that when firms initially substitute into cash, they also substitute away from capital and labour by diverting some of the resources previously used for producing output. The fall in output relative to consumption puts downward pressure on the real wage. In the new equilibrium, a lower real wage results in lower consumption and slightly higher equilibrium labour. Both factors contribute to a drop in consumer welfare of approximately 0.1%. In the end, the change in welfare happens because of an interesting interplay between real and financial policies, and this experiment emphasizes the importance of the effect of tax incentives on a firm's optimal substitution between debt and capital.

5.7. Model extensions and alternatives

In this section, we consider three different extensions of the baseline model in Section 5. The goal is to understand whether and how the results in Figures 5 and 6 depend upon the exact form of financial frictions in the model. We also consider a separate class of models.

First, we examine a model in which debt is constrained by collateral. Specifically, we replace equations (12) and (13) with a condition that prevents debt from surpassing a collateralizable fraction of the capital stock, as follows:

$$p \leq (1 - \xi)(1 - \delta)k \quad (18)$$

In (18), we have redefined the parameter ξ to be the fraction of the capital stock that cannot be used as collateral. In this case, the firm never defaults, and the interest rate on debt is the risk-free rate and thus invariant to firm size or leverage choices.

Second, we consider an alternative model in which we allow debt to be risky by reinstating equations (12) and (13), but we allow the firm to issue costly equity. Specifically, we replace the constraint in (10), with a condition that redefines negative cash flows to shareholders as $e(k, p, n, k', p', z, \tau)(1 + \lambda)$, in which λ is a linear cost of equity issuance. Because we have no data on private placement issuance costs for small firms, especially the implicit costs for those that never issue, we set $\lambda = 0.11$ ([Hennessy and Whited, 2007](#)).

Third, we consider an extension where there is no external equity issuance, and default occurs only when firm value equals zero. Specifically, we replace equation (12) with:

$$\Pi(k, p, z, \tau) < 0 \quad (19)$$

We estimate these three alternative models using our sample of small firms. The results are in [Online Appendix Table B20](#). In all three cases, we find parameter estimates comparable to those from our baseline estimation. In [Online Appendix Figure B4](#), we plot the policy functions from these three alternative models, finding in all cases a positive relation between the productivity shock and leverage and a negative relation between the tax shock and leverage, as in our baseline results in Figure 5. For the model with an equity issuance cost, a linear cost of 0.11 results in rare equity issuance, so it is not surprising that the behaviour of a firm that rarely issues equity is similar to that of a firm that issues no equity. In the model with a collateral constraint, as

explained in [Rampini and Viswanathan \(2013\)](#) and [Li *et al.* \(2016\)](#), the cost of debt is lost future financial flexibility, which manifests itself in the model as the expectation of the future Lagrange multiplier on the constraint in (10). Because this expectation depends on optimal policies chosen by the firm, it is naturally tax-sensitive, with higher taxes making the loss of financial flexibility more costly. In the model with solvency default, although the default threshold is less stringent, the effects of tax-induced substitution between debt and capital still operate, and the effect of taxes on the default premium is similar.

Next, we consider an extension of our model that addresses the differences in effective and enactment dates in our data. The baseline version of the model contains taxes that become effective immediately, yet in our data, 37% of the tax acts take effect at least one year after the legislation passes. To incorporate delayed taxation into our model, we replace (4) with a process that contains a news shock as in [Jaimovich and Rebelo \(2009\)](#):

$$\tau' = \rho_\tau \tau + \sigma_\tau u' + \sigma_v v, \quad v, u' \sim \mathcal{N}(0, 1). \quad (20)$$

In this process, a realization of v today directly affects tomorrow's tax rate, τ' .

To understand how leverage responds to the news shock, v , we set the variance and serial correlation of the process for τ equal to their baseline values. Because we do not have a long enough time series on state corporate taxes to estimate a news shock variance, we set the relative variances of u' and v so that conditional on $|u'| > |v|$, the mean of $\Delta\tau'$ is 1.15 percentage points larger than the mean of τ' conditional on $|u'| \leq |v|$, which is the difference in our data between the average anticipated tax change and the average unanticipated tax change. This process implies that the ratio, $\sigma_v^2/(\sigma_u^2 + \sigma_v^2)$, is 0.579. We then reestimate this augmented model with the results in [Online Appendix Table B20](#). This extension is important because we now use a tax elasticity that reflects anticipation to estimate a model that embodies anticipation. We plot the policy functions for leverage in Panel D of [Online Appendix Figure B4](#). We find that a higher news shock realization results in lower leverage, as is the case for a tax shock in our baseline model estimated on the sample of small firms.

Next, we consider the response of leverage to taxes in a different class of dynamic models: those based on the frameworks in [Fischer *et al.* \(1989\)](#) and [Leland \(1994\)](#). In these contingent-claims models, the value of debt depends on the value of the default put option, and the optimal choice of leverage reflects default. Specifically, we consider the dynamic version of the model in Section III of [Goldstein *et al.* \(2001\)](#). We solve the model for several different underlying parameter values and plot the relation between optimal leverage and the tax rate in [Online Appendix Figure B5](#). Five main parameters govern the optimal choice of leverage in this model: the risk-free interest rate, the debt issuance cost, the firm's default cost, and the drift and volatility of profit, which is characterized by a nonstationary diffusion process. In each of Panels A–E in [Online Appendix Figure B5](#), we plot the relation between optimal leverage and the tax rate for high and low values of each of these parameters.

We always find that a higher tax rate results in higher optimal leverage. Three main features of this model differ from ours and underlie the result. First, in this class of models, shareholders can always raise equity costlessly. Thus, although taxes affect default thresholds, they always raise leverage because the assumption of costless equity implies that financial distress happens rarely, so the effect of taxes on the default put is small relative to the direct tax advantage of debt. Second, in this class of models, there is no capital, so there is no substitution between capital and a storage technology, that is, debt. Third, firms have no control over their distance to default, while in our model, firms can optimally choose to be near the default threshold, so the effect of taxes on the cost of default matters for these firms.

5.8. *External validity*

One of the main implications of our model is testable: risky loan rates should rise with taxes because firms are more likely to default when taxes are higher. While the interest rate information in the Y-14 data is too sparse and noisy to test this hypothesis, the Y-14 collection contains information on banks' estimates of firms' default probabilities. Therefore, we can test the hypothesis that these default probabilities move with tax rates. We limit the sample to private firms and exclude observations corresponding to defaulted firms and to those with default probability estimates outside the (0, 1) interval. We use a sample from the first quarter of 2012 through the fourth quarter of 2018. Requiring the sample to end in 2017, as for our firm balance sheet sample, would give us 3 years of data for most firms because the coverage of the risk metrics variables does not become complete until the end of 2014 (Beyhaghi *et al.*, 2024). Unlike our firm balance sheet sample, the risk metrics panel is quarterly. Finally, we do not require firms to have available financial statement information, which results in a substantially larger sample of firm quarters.

We use these data to estimate the evolution of default probabilities around corporate income tax cuts in an event study. We include the following controls: borrower and 4-digit NAICS by year-quarter fixed effects, state individual income tax cut and hike indicators, a set of tax base and tax credit controls, and lagged changes and levels of county GDP, employment, per-capita income, and population. We consider various bins of total commitment amount sizes: \$10–\$100, $\geq \$10$, \$10–\$50, \$20–\$100, \$50–\$100, $\geq \$100$, and $\geq \$50$ million. Ideally, we would compare small and large firms, but we do not have balance sheet information for most observations in the risk metrics panel. However, Sufi (2009) and Brown *et al.* (2021) show that conditional on firms having credit line financing, credit line size is about 20–25% of firm assets, while Rauh and Sufi (2010) show that term loans and credit lines each constitute roughly half of firms' total bank debt. As such, firms in the \$10–\$100 or \$10–\$50 million buckets are likely to have between \$20–\$400 and \$20–\$200 million in assets, respectively, and be small firms.

Table 3 shows the evolution of default probabilities around state corporate income tax cuts. Columns (1)–(4) show that these probabilities decline significantly in the year of enactment or one year after enactment by 20–30 basis points. These magnitudes represent an approximate 10% fall from average default rates that average around 2%. (See Online Appendix Table B21.) This decline peaks 2 years after enactment and remains significant through 3 years after enactment. These results are driven by firms that have total commitments below \$50 or \$100 million. By contrast, we do not observe much change in default probabilities around corporate income tax cuts for the larger loan sizes.

Next, we examine a less direct measure of default probability from the SNC Program: one based on banks' internal credit risk assessment and another based on supervisory assessment. These variables are one when at least one of the firm's loan commitments is of non-pass credit quality and zero otherwise. A non-pass rating is associated with a significantly higher probability of banks setting aside loan loss reserves and charge-offs (Ivanov *et al.*, 2023). In Online Appendix Table B19, we show the event study results for each measure of firm risk for both tax cuts and hikes. The first two columns show a 1.7–2.2% point fall in the probability of a non-pass rating accompanying tax cuts. These drops persist for roughly 3 years following enactment and are consistent across the two measures. In contrast, tax hikes are not associated with any change in firm risk. From the results in Table 3 and Online Appendix Table B19, we conclude that taxes affect firm default likelihood, so these results offer a useful check on the mechanism in the model whereby taxes affect borrowing terms.

TABLE 3
Probabilities of default and corporate tax cuts

	Probability of Default						
	10–100	10–	10–50	20–100	50–100	100–	50–
Event Year = –2	–0.0004 (0.0007)	–0.0002 (0.0006)	–0.0003 (0.0008)	–0.0008 (0.0010)	–0.0007 (0.0020)	0.0008 (0.0024)	0.0004 (0.0016)
Event Year = –1	–0.0004 (0.0014)	–0.0003 (0.0014)	–0.0003 (0.0012)	–0.0019 (0.0018)	0.0006 (0.0027)	0.0053 (0.0033)	0.0018 (0.0022)
Event Year = 0	–0.0024 (0.0015)	–0.0023* (0.0013)	–0.0028** (0.0011)	–0.0034 (0.0020)	0.0017 (0.0048)	0.0012 (0.0030)	0.0013 (0.0023)
Event Year = +1	–0.0032** (0.0014)	–0.0029** (0.0012)	–0.0032** (0.0014)	–0.0031* (0.0018)	–0.0048 (0.0047)	–0.0011 (0.0037)	–0.0007 (0.0032)
Event Year = +2	–0.0034*** (0.0011)	–0.0033*** (0.0010)	–0.0038*** (0.0009)	–0.0034** (0.0016)	–0.0030 (0.0048)	0.0003 (0.0028)	0.0005 (0.0027)
Event Year = +3	–0.0013* (0.0008)	–0.0015** (0.0007)	–0.0014* (0.0008)	–0.0008 (0.0009)	–0.0006 (0.0023)	–0.0003 (0.0020)	–0.0012 (0.0017)
Event Year = ≥ +4	–0.0009 (0.0007)	–0.0011* (0.0006)	–0.0009 (0.0008)	–0.0000 (0.0009)	0.0019 (0.0023)	–0.0063* (0.0032)	–0.0005 (0.0017)
Observations	172,084	195,589	145,149	94,336	22,590	12,865	39,430
R^2	0.711	0.701	0.727	0.726	0.774	0.759	0.726
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
TB Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Economy Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents results from firm-year panel regressions of the evolution of firm probabilities of default around the enactment of state corporate income tax cuts. The sample runs from 2012 through 2019 and is restricted to private companies. The dependent variable in all specifications is the current-quarter probability of default estimate for each firm in the sample. Y-14 banks report the probability of default estimates as part of the Federal Reserve's Y-14 Collection. All specifications include both borrower and 4-digit NAICS by year-quarter fixed effects. The regressors are state corporate income tax hike indicators, state individual income tax cuts and hike indicators, a set of tax base and tax credit controls, as well as lagged changes and levels of county GDP, employment, per-capita income, and population. Columns (1) through (7) present results for borrowers with \$10–\$100, ≥\$10, \$10–\$50, \$20–\$100, \$50–\$100, ≥\$100, and ≥\$50 million in total commitments. The standard errors are double clustered at the state and 4-digit NAICS by year-quarter level. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

6. CONCLUSION

Using comprehensive samples of both private and public companies and relying on simple event study techniques, we study the evolution of corporate borrowing around changes in state corporate income taxes since the 1990s. In contrast to much of the prior literature, we show that for small private firms, corporate taxes, on average, depress their leverage, that these effects last many years, and that companies adjust capital structure immediately upon the enactment of changes in state corporate tax policy. Interestingly, the leverage of large private firms does not change in response to tax legislation.

We rationalize these empirical results in a dynamic equilibrium model. The model is intentionally standard, with a tax benefit of debt baked in. Firms make optimal leverage, investment, and hiring decisions in an infinite-horizon framework, while facing restricted external equity finance. We estimate the model parameters on different samples of firms and show that optimal leverage can either rise or fall with the tax rate. For the model estimated on our sample of small firms, the tax benefit is offset by a tax-sensitive cost that reflects the negative effect of taxes on firm health and, thus, a positive effect on equilibrium credit spreads. The taxation of profit also offsets the tax benefit, so a tax hike lowers optimal capital and, therefore, the leverage

needed to finance it. In contrast, for the model estimated on our sample of large firms, leverage exhibits a traditional positive relation to the corporate tax rate. Because this second parameterization of the model features sluggish investment, the investment response to a tax cut is muted, the accompanying need to borrow is also muted, and the fall in the tax benefit guides a fall in leverage.

We also use our model to examine the welfare effects of the tax benefit of debt. We find that welfare falls when the tax benefit is removed from the model. Without this extra incentive to finance investment, firms hold too much cash and too little capital. The equilibrium result is more equilibrium hours worked, lower consumption, and lower welfare. This result has a second-best flavour because the tax benefit incentivizes firms to get closer to a frictionless optimal capital stock when they face constraints on external equity finance.

Our main result of a negative sensitivity of leverage to taxes is for small private U.S. firms. While these firms do not constitute the majority of U.S. output, most firms outside of the U.S. are both small and private. Thus, our results have implications for large classes of firms in both developed and developing economies, and it is worth asking whether the same sets of frictions and trade-offs apply to firms in other parts of the world.

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Supplementary Data

Supplementary data are available at [Review of Economic Studies](#) online.

Data Availability

The code and non-proprietary data underlying this research are available on Zenodo at <https://doi.org/10.5281/zenodo.15991628>.

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